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False Reasoning, True Consequences: Fallacy Detection in Italian Social Media with Multi-Task Learning

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Garbage In, Garbage Out
- George Fuechsel, 1960s

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Abstract

In an era where a tweet can shape public opinion and fallacious reasoning spreads virally, automatically detecting fallacies has become both a computational challenge and a social necessity. This thesis delves into the field of Automatic Fallacy Detection (AFD), presenting our participation as team TiGRO (Turin in GROningen) at FadeIT: the first shared task on AFD in Italian social media texts, organized within the EVALITA 2026 evaluation campaign.

The task is based on the FAINA dataset, consisting of 1,440 Italian Twitter/X posts on migration, climate change, and public health, and annotated at the span level with 20 fallacy types by two expert annotators following a perspectivist approach to allow genuine disagreement. We participated in both subtasks: post-level fallacy classification (Subtask A) and span-level fallacy identification (Subtask B).

Our core approach is *E pluribus unum* ("from many, one"), using multi-task learning to leverage signal sharing. Each fallacy class is treated as an independent binary task, jointly trained over a single shared encoder (ALBERTo or mmBERT). Our best model extends this idea across both subtasks, jointly fine-tuning 40 parallel tasks that combine post-level classification and token-level span identification for all 20 fallacy types simultaneously.

TiGRO achieved first place in Subtask A (micro-F1: 56.39) and first place under strict evaluation in Subtask B (micro-F1: 42.13), with fourth place under soft evaluation (micro-F1: 44.98). Despite these successes, we provide an extensive per-class analysis, highlighting that performance is heavily driven by class frequency and surface lexical cues, as confirmed by comparison with a simple SVM baseline. We then critically discuss the limitations of micro-F1 as an evaluation metric under severe class imbalance, as well as structural constraints of the FAINA dataset, such as its skewed distribution, insufficient annotator diversity, and annotation inconsistency.

Our findings provide an empirical grounding for well-known challenges in AFD and motivate a broader discussion of future directions for robust approaches, including improved dataset construction, multi-domain training, and hybrid architectures combining encoder-based models with large language models.

Declaration of originality

I declare to be responsible for the content I'm presenting in order to obtain the final degree, not to have plagiarized in all or part of, the work produced by others and having cited original sources in consistent way with current plagiarism regulations and copyright. I am also aware that in case of false declaration, I could incur in law penalties and my admission to final exam could be denied

Declaration on Generative AI

During the preparation of this work, the authors used ChatGPT, Claude and Grammarly in order to: Grammar and spelling check, Text Translation, Improve writing style. After using these services, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

Content Note. This thesis includes examples of social media posts drawn from the FAINA dataset that may contain offensive, politically charged, or otherwise disturbing language for some readers. These examples are reproduced solely for scientific analysis.

Contents

Introduction

Fallacies are arguments that *appear* to be valid but are not. While their presence in public debate is not new, the rise of social media has dramatically amplified their reach and impact, playing a key role in shaping misleading narratives that threaten democratic processes. In this context, *Automatic Fallacy Detection* (AFD) has recently attracted increasing attention as a means to counter *misinformation*.

This thesis presents our participation as team **TiGRO** (Turin in GROningen) [1] at the **FadeIT** [2]: the first shared task on AFD in Italian social media texts, organized within the EVALITA 2026 evaluation campaign [3].

The task involves identifying 20 fallacy types in Italian social media posts at both the post level (Subtask A) and the span level (Subtask B), and is based on the **FAINA** dataset [4] composed of 1,440 Italian Twitter/X posts annotated by two equally valid annotators, following a perspectivist framework.

Our core approach, captured by the motto *E pluribus unum* ("from many, one"), is based on the joint modelling of both subtasks through **multi-task learning**: rather than treating a multi-class problem, we decompose it into multiple independent binary tasks. Concretely, our best-performing systems fine-tune a single shared encoder (ALBERTo or mmBERT) on 40 parallel tasks: 20 for post-level classification and 20 span-level identification, encouraging signal sharing across fallacy classes and levels of granularity. This architecture led to strong results, achieving first place for Subtask A and fourth place for Subtask B.

The thesis is organized as follows.

Chapter 1 introduces the study of fallacies, tracing the evolution of the field from Aristotle to the modern era. It highlights the conceptual complexity that makes fallacies difficult to define and classify, and emphasizes their relevance in the context of misinformation and post-truth public discourse. The chapter also presents the 20 fallacy classes used in this work, summarized in Appendix A for ease of reference.

Chapter 2 provides the technical background underlying our system, covering everything needed to understand this work: the Transformer architecture, pre-trained encoder models, and feature attribution methods.

Chapter 3 presents the FadeIT shared task and describes the FAINA dataset, including the perspectivist annotation approach that distinguishes it from related works. Finally, it provides an overview of other participant systems and evaluation.

Chapter 4 describes the TiGRO systems, covering encoder choice, handling of dual annotations, multi-task architecture, and training setup. It then presents results for both subtasks with per-class breakdowns, comparative baselines, and discussions on performance. It also presents experimental approaches explored.

Chapter 5 reflects critically on the implications of our results and the limitations of the FAINA dataset, then discusses what would be needed to move the field forward, situating our findings within the broader challenges facing AFD.

Chapter 1

Fallacies

This chapter introduces the reader to the complex domain of *fallacy* study. Section 1.1 provides a brief overview of the concept, highlighting the lack of a universally accepted definition. Section 1.2 traces the field historically. Section 1.3 surveys the modern landscape, covering formal and informal approaches and the linguistic and psychological dimensions of fallacious reasoning. Section 1.4 discusses why the topic matters, particularly in the context of misinformation and social media in the post-truth era. Finally, Section 1.5 describes the fallacy classification adopted in this work, drawn from the FAINA dataset [4] and used for the FadeIT Shared Task [2].

1.1 A non-definition of fallacy

Common understanding In popular discourse, a fallacy is often understood as a false but popular belief, frequently based on stereotypes. The Cambridge Dictionary defines a fallacy as *"an idea that a lot of people think is true but is in fact false"*. For example, *"Women are worse drivers than men"*, which is obviously not true.

However, academic writers generally prefer to consider a fallacy as a deceptively erroneous argument. Such arguments can be constructed either intentionally, to persuade or manipulate through deception, or unintentionally, due to negligence, ignorance, or cognitive and social biases [5, 6, 7, 8].

In this sense, a fallacy is a fallacious argument. A simple false statement is not considered a fallacy if it does not constitute an argument; that is, if it does not contain one or more premises leading to a conclusion. For example, consider the classic logical fallacy of affirming the consequent: *"If it rains, the streets get wet. The streets are wet; therefore, it rained"*. The conclusion might seem reasonable at first glance, but the argument is invalid because the streets could be wet for other reasons. Importantly, we are not concerned with the truth of the premises or the conclusion, but rather with understanding what makes the argument appear stronger or more convincing than it actually is.

Lack of Definition It must be emphasized that there is currently no widely accepted theory of fallacies. The literature is extensive and intricate, with many overlapping dimensions and approaches, which has led to considerable confusion. There is not even a universally accepted definition, and some scholars might even contest the brief explanation given above.

Without a formal definition, it is difficult to determine precisely what counts as a fallacy and what does not, and even more challenging to classify the different types of fallacies. Even if multiple classifications have been proposed, it is often argued that fallacies cannot be systematically classified [5].

De Morgan in Formal Logic says:

> *"There is no such thing as a classification of the ways in which men may arrive at an error; it is much to be doubted whether there ever can be."* [9]

H.W.B. Joseph in Introduction to Logic says:

> *"Truth may have its norms, but error is infinite in its aberrations, and they cannot be digested in any classification."* [10]

Cohen and Nagel in Introduction to Logic and Scientific Method say:

> *"It would be impossible to enumerate all the abuses of logical principles occurring in the diverse matters in which men are interested."* [11]

1.2 A brief history of fallacies

The study of fallacies can be traced back to Aristotle (384–322 BC), particularly to his works on dialectical argumentation and rhetoric, in which he discusses the art of persuasion in dialogue. In *"On Sophistical Refutations"*, he listed and examined thirteen types of fallacies, laying the foundation for the field, which nevertheless remained largely stagnant for nearly two millennia. As Henry Aldrich (1647-1710) noted: *"since it satisfied Aristotle, it has satisfied all later logicians"* [12]. The few proposals during this period mostly involved rearranging existing raw material, and fallacies remained largely relegated to appendices in logic textbooks.

A pivotal moment occurred with Charles Hamblin's book *Fallacies* (1970) [5], which criticized the standard dogmatic treatment of fallacies and successfully renewed interest in fallacy theory. Hamblin provided one of the most cited definitions of fallacy: *"an argument that seems to be valid but is not."*. While widely accepted, this definition also raised various concerns, leading to the development of multiple, intricate, and highly debated approaches in the post-Hamblin era. These include formal approaches, pragma-dialectical approaches (van Eemeren and Grootendorst [13, 14, 15, 16]), epistemological approaches (Siegel and Biro [17, 18]), pragmatic approaches (Douglas Walton [19]), multi-logical approaches (John Woods and Douglas Walton [20]), and psychological approaches [21, 22, 23].

1.3 The modern approach

Formal vs Informal Fallacies In modern times, it is common to distinguish between *formal* and *informal* fallacies [24]. The former involves errors in the structure of the argument, whereas in the latter, the problem lies in the content, context, or language. More specifically, formal fallacies consist of just a few instances of incorrect logical form, such as the undistributed middle or denying the antecedent. In contrast, informal fallacies consist of a vast heterogeneous group of different types that have not yet found an adequate formal representation, which would allow them to be modelled systematically.

This distinction reflects deeper issues: it is unclear whether logic should aim to provide explicit definitions of invalid arguments, or whether it should restrict itself to defining validity and consider invalid any argument that fails to meet those standards. More fundamentally, it remains an open question whether fallacies should be studied within formal logic at all, and whether it is possible or desirable.

These concerns are addressed in Gerald Massey's *Asymmetry Thesis*, one of the most significant critiques of the formal approach to fallacy theory. According to Massey, while valid arguments can be demonstrated as such through formal proof, there is no equally robust and systematic method for proving that a natural-language argument is invalid. From this asymmetry, he concludes that a general theory of fallacies cannot be grounded solely in formal logic [25].

The informal approach allows the analysis of fallacies in detail without committing to a rigid formal language, instead acknowledging their heterogeneous nature and their intersections with logic, linguistics, philosophy, and psychology.

The Linguistic Dimension The difficulty in addressing informal fallacies largely stems from their grounding in natural language and one of its defining characteristics: ambiguity. Every informal fallacy has to do, to some extent, with the language in which an argument is expressed [26].

One of the most discussed examples, dating back to Aristotle, is *equivocation*, where a term with multiple meanings is used in different senses within the same argument, creating a misleading appearance of validity. Such phenomena are difficult to capture within a formal language, which, by definition, is not ambiguous.

This linguistic complexity explains why fallacies have also been studied from perspectives beyond formal logic. Fallacies are behavioral symptoms of our irrational thinking, and they are an important topic of study because they reveal something about human nature. Fallacies are closely related to the way we express ourselves: everyday discourse is not composed of lists of unequivocal facts, but of flexible and creative language. During a debate, the right word at the right moment can strengthen our position, although having no real impact on the formal logic of our reasoning.

The Psychological Dimension Hamblin's definition makes clear that a fallacy is not merely an invalid argument; it is an argument that *appears* to be valid but is not. The notion of appearance, also referred to as the *"appearance condition"*, captures a central feature of fallacies. However, this condition has been criticized for its subjectivity and lack of formal precision, leading some logicians to argue that fallacies fall outside the proper domain of logic. For instance, Massey affirms that fallacies are primarily a psychological rather than a logical phenomenon [25]. On this view, fallacies are not distortions of logic, but distortions of patterns of thinking, and the focus shifts from their formal structure to the cognitive mechanisms that make them persuasive.

This shift suggests a natural point of contact between fallacy theory and the study of cognitive biases, understood as systematic patterns of deviation in human reasoning that can lead to faulty assessment. Like fallacy theory, research on cognitive bias is complex and multifaceted. Several studies have attempted to draw connections between specific fallacies and underlying cognitive biases, as they seem to fundamentally investigate the same topic: *errors* in reasoning [21, 22, 23].

However, while the study of bias may help explain why people commit certain errors, the study of fallacies aims to characterize the nature of those errors themselves. Thus, both approaches are significant, and the relationship between them remains an open and fruitful area of investigation.

Pragmatic Approaches The formal approach has not been entirely abandoned. In recent decades, Douglas Walton and John Woods have made significant contributions by advocating a pragmatic and pluralistic methodology [19, 20]. They acknowledge the complexity of fallacies and support the appearance condition, while arguing that traditional formal logic is too coarse-grained to capture the distinctive features of individual fallacies.

Their approach combines elements of formal and informal analysis: they examine each fallacy in its specific argumentative context, but they preserve a certain formal rigor, employing the most fitting logical tools depending on the case.

1.4 Importance of fallacies

Following our discussion, the study of fallacies is highly relevant from a philosophical perspective. The ability to distinguish a sound argumentative structure from a fallacious one is fundamental for constructing reliable reasoning in the pursuit of truth. Moreover, the analysis of error itself represents a central and intellectually valuable topic within logic and epistemology.

However, fallacies are not merely theoretical concerns confined only to academia. Understanding the fundamentals of reasoning shapes our individual capacity for critical thinking and, as functioning members of a democracy, this competence has significant social and political implications. Indeed, argumentation and dialogue are essential components of democratic life and peaceful coexistence among individuals, yet when public discourse is dominated by fallacious reasoning, these foundations are weakened, and consequences can be profound, affecting collective decision-making, public trust, and the quality of democratic institutions.

Fallacies are widespread across highly contested domains and are particularly harmful in political debate, where they may be strategically employed to persuade audiences and gain support, thus poisoning public discourse and contributing to the spread of *misinformation* and fake news.

This issue is amplified by the current information ecosystem, characterized by the rapid diffusion of content through digital media. Public debate no longer occurs exclusively in institutional settings, but it has mainly moved to social media platforms, taking place in any comment section. Nowadays, almost every topic is treated as a debate, and every choice is seen as taking a position. This environment traps individuals in ideological bubbles, polarizing society into two opposing factions: "*us*" versus "*them*". These dynamics are further reinforced by platform algorithms and content creators, who benefit from the high levels of engagement and interaction generated by these discussions.

In such settings, people are frequently exposed to information that reinforces their pre-existing beliefs. As a result, they may defend their side vigorously, even when doing so involves fallacious reasoning, whether intentional or unintentional. At the same time, their critical defenses can weaken, making them vulnerable and more likely to blindly accept claims from their own side without proper evaluation.

Such dynamics have a real-world impact on society and contribute to lowering the standards of public debate in what is often described as a post-truth era, in which *"objective facts are less influential in shaping public opinion than appeals to emotion and personal belief"*[1].

The consequences of this phenomenon are evident in recent political and social events. For instance, a survey conducted in July 2023 found that almost 40% of the US public, and almost 70% of Republicans, negated the legitimacy of the 2020 presidential election outcome [27]. Similarly, previous research has highlighted the role of fallacious reasoning in the Brexit referendum [28] and in the spread of misinformation during the COVID-19 pandemic [29]. These examples illustrate that the danger of misinformation is not only in the factual content of information, but often in the *misleading reasoning* and argumentation used to present it.

In other words, what makes fake news particularly harmful is not always its falsity, but the way arguments are framed using fallacious techniques, such as *false analogies*, *hasty generalizations*, and c*herry-picking*. This type of content, generally referred to as misinformation, may not be deliberately intended to deceive, but it still distorts understanding and influences public opinion [30]. Supporting this, a study on the COVID-19 "infodemic" [31] found that 59% of fake news did not contain fabricated content, but instead reconfigured existing information in misleading ways. The proliferation of such content through social media, the primary news source for many vulnerable populations, highlights the urgent need to improve fact-checking, inform policy-making, and promote digital literacy and critical thinking [32].

The urgency of this problem, combined with recent advances in Natural Language Processing (NLP), has led to growing interest in *Automatic Fallacy Detection* (AFD). In recent years, the field has expanded significantly, with the introduction of annotated datasets across domains such as politics and social media [33, 34, 35, 36, 4], the development of large language model–based approaches [37, 38, 39, 40, 41], and the organization of shared tasks on fallacy detection [42, 2]. These efforts aim to provide computational tools to identify fallacious reasoning and mitigate its harmful impact on public discourse and society.

1.5 Fallacy Classification

Fallacies are numerous and varied, with modern lists identifying hundreds of distinct types; for instance, the Internet Encyclopedia of Philosophy documents over 200 common formal and informal fallacies [43]. However, as noted in Section 1.1, there is currently no universally accepted theory or classification of fallacies, and although many attempts have been made to create classification systems, none have achieved widespread consensus.

This poses a challenging problem for AFD, which has renewed attention on this topic by following a more pragmatic approach that is not grounded in theoretical implications but is instead driven by the goal of solving the task.

However, in the absence of greater clarity, several studies have proposed new classification schemas or attempted to harmonize existing ones, which has resulted in a convoluted literature with differing frames of reference [33, 34, 35, 36, 4]. Various works define different numbers of classes, hierarchies, and fallacy types, often using

[1] Oxford Dictionary

datasets from distinct domains and contexts. Distinguishing between these classes is fundamental for developing automatic classifiers, and the adoption of a standard classification would allow for a fair comparison between different systems.

While it is hoped that the task definitions will eventually converge, this problem is far from resolved, and trying to produce a fully satisfactory classification is beyond the scope of this work. Therefore, in this section, we limit ourselves to describing the classes that were used in this work for the Shared Task FadeIT [2], obtained from the FAINA dataset introduced by Ramponi et al. [4] (see Section 3.3). This classification is primarily driven by pragmatic considerations for automatic fallacy detection in social media, without strict adherence to any particular theoretical framework.

A total of 20 fallacy types were identified by harmonizing definitions and names from Da San Martino et al. [44], Musi et al. [32], among others. Following similar efforts [8, 45], these fallacy types are further organized into a taxonomy, shown in Figure 1.1, which groups them into macro-categories: *Insufficient Proof* (e.g., Evading the Burden of Proof, Vagueness), *Simplification* (e.g., Hasty Generalization, Slippery Slope, False Dilemma), and *Distraction* (e.g., Loaded Language, Appeal to Emotion, Ad Hominem). However, this taxonomy is mainly intended for evaluation purposes and does not claim rigorous grounding.

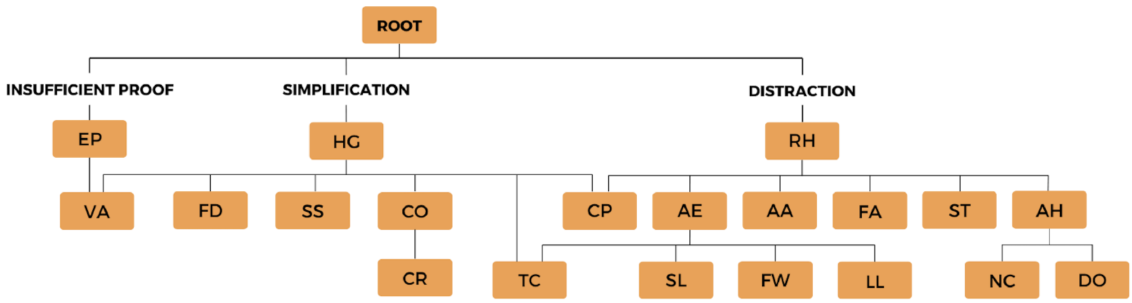


Figure 1.1: Taxonomy of fallacy types in FAINA dataset Ramponi et al. [4]

To provide a clear and practical overview, we now describe each fallacy type individually. For each fallacy, we provide a definition along with a toy example. This structured presentation allows readers to understand the distinctions between classes and provides the necessary context for subsequent discussions. A complete table summarizing all fallacies, their definitions, and FAINA dataset examples can be found in Appendix A.

List of fallacies

Ad Hominem (AH) Latin for "*to the person*". This fallacy involves a personal attack on an individual or a group that deviates from the main thesis. It operates as a diversionary tactic, incorrectly assuming that a person's character or background is a direct indicator of the truthfulness of their logical claims. It comprises *Tu Quoque* (i.e., when attacking the contradiction between the opponent's claim and action), *Circumstantial* (i.e., when personal interests are questioned), and *Abusive* (i.e., when directly attacking the opponent person) [46]. Example: "We can't trust his economic plan because he cheated on his wife!"

Appeal to Authority (AA) When appealing to an authority or a group consensus (*Ad populum*) to support their thesis without further evidence. This technique exploits the audience's psychological tendency to transfer trust from a person's status to their opinions, even when it is irrelevant, inappropriate, or unsupported [33].
Example: "This cereal is healthy because my favorite athlete eats it every day."

Appeal to Emotion (AE) This technique involves the use of negative or positive personal emotions to influence the audience. By triggering affective responses, the arguer attempts to bypass the recipient's rational scrutiny and the need for evidentiary support [44]. is often manifested on social media through a heavy use of emojis. It can be further divided into subcategories based on the emotion triggered: *Appeal to Pity, Appeal to Anger, Appeal to Fear, Appeal to Ridicule, Appeal to Positive Emotion* [36].
Example: "If you don't support this policy, you are putting your children's lives at risk."

Causal Oversimplification (CO) This fallacy involves a simplified and fallacious causal relation. It often transfers blame for a multifaceted problem to a single cause, providing a superficial explanation that ignores real-world complexities. It includes *False cause* (i.e., when the wrong cause is attributed to an effect) and *post hoc* (i.e., when correlation is presented as causation) [32]
Example: "Every sunrise, the rooster makes a sound. So it is the rooster that makes the sun rise."

Cherry Picking (CP) This consists of choosing evidence to support a thesis while ignoring any contrary evidence. It distorts reality by presenting a biased selection of data points as if they represented the complete truth [32]. Unlike *Hasty Generalization* (i.e., generalizing from a too small sample), *Cherry Picking* involves selecting only the evidence that supports a pre-existing conclusion.
Example: "This drug works because two patients recovered (even though fifty others did not)."

Circular Reasoning (CR) An error of circularity where the end of an argument comes back to the beginning without having proven itself. This is considered a pragmatic defect where the premises are just as much in need of proof as the conclusion, resulting in self-authentication [35].
Example: "I am in charge because I am the leader."

Doubt (DO) Used to intentionally question the credibility of someone or something without directly refuting their argument. Unlike healthy skepticism, it functions as a form of *ad hominem* attack intended to instill uncertainty in the opponent's position [44].
Example: "Is he ready to be the Mayor?"

Evading the Burden of Proof (EP) A thesis is advanced without any support as if it were self-evident, meaning arguments are missing. It also contains *Shifting the burden of proof* (i.e., when demanding the opponent justify the opposite of the claim). This violates the logical standard of "*onus probandi*", which dictates that the claimant is responsible for justifying their claim [32].
Example: "Obviously, violent video games cause aggression among teenagers."

False Analogy (FA) Two different things are placed on the same level because they are supposed to share similar aspects. The error lies in a categorical mismatch where superficial similarities are used to draw a conclusion that fails to account for relevant differences [32].
Example: "Murder is immoral. Therefore, abortion is immoral."

False Dilemma (FD) This fallacy presents only two options when, in reality, a wider range of possibilities exists. It creates an artificial dichotomy that pressures the audience into making a hasty decision, often by framing the alternative choice as an undesirable or disastrous outcome [35].
Example: "If you don't major in computer science, you'll never get a good job."

Flag Waving (FW) Used to intentionally play on a sense of belonging to a certain country, community, or ideology to support an argument. It is a specific type of *Appeal to Emotion*, similar to *Slogan*, where the arguer attempts to use symbols to trigger non-conscious feelings of unity and bypass logical thought. [44].
Example: "If you are a true patriot, vote yes!"

Hasty Generalization (HG) Generalization drawn from a sample too small or unrepresentative. It involves a logical leap from a few particulars to a universal rule, often leading to inaccurate stereotypes [32, 35].
Example: "I met two aggressive dogs, so all dogs must be aggressive."

Loaded Language (LL) The use of language with strong emotional implications to influence the audience. It is a specific type of *Appeal to Emotion*, in which strongly connoted terms carry an "evaluative meaning" as a hidden layer in addition to their descriptive content, suggesting judgment without evidence, and subtly shaping our perception [44].
Example: "This reckless government is gambling with our future."

Name Calling or Labelling (NC) Labeling someone or something positively or negatively to influence the audience. It is a type of *Abusive Ad Hominem* that aims to construct a bad image of a target using names the audience fears or hates, thereby devaluing the speaker's argument through stigma [44]. It comprises *Reduction ad Hitlerum* [44] also known as *Guilt by Association* [36] (i.e., when the opponent is discredited based on their association with another person, group, or idea that is viewed negatively).
Example: "Don't listen to that green activist; he is just a radical tree-hugger."

Red Herring (RH) The argument diverts attention to irrelevant issues. This strategy leads the audience away from the core dispute and toward a claim that may be true but is not relevant to the original conclusion, often to avoid accountability. It includes *Appeal to worse problems* (i.e., when the argument is eluded because there is something worse that should be addressed), *Appeal to tradition* (i.e., when something is justified because it has always been done that way), and *Appeal to nature* (i.e., assuming that something is good simply because it is natural) [47, 36].
Example: "Why worry about space exploration when we have homeless people in our country?"

Slippery Slope (SS) Implies that an exaggerated consequence could result from an action. It relies on the "domino effect" to discourage change by claiming that a small first step will inevitably trigger an uncontrollable negative chain reaction, ultimately leading to an exaggerated tragedy [33]
Example: "If you don't study, you'll fail, never get a job, and become homeless."

Slogan (SL) Brief and striking phrase used to provoke excitement. Slogans simplify complex social or political issues into catchy rallying cries that promote a specific ideological framing. It is a specific type of *Appeal to Emotion*, similar to *Flag Waving*, often manifested as hashtags in social media [33].
Example: "Make America Great Again!"

Strawman (ST) Distorting an opponent's argument in order to make it easier to attack. Instead of addressing the actual claim, the arguer misrepresents it, refutes this weakened or exaggerated version, and then concludes that the original position has been defeated. By attacking a caricatured interpretation of the claim, the arguer creates an illusion of victory while avoiding engagement with the real issue [32].
Example: "He supports legal cannabis, so he wants everybody to be junkies."

Thought Terminating Cliches (TC) Short and generic phrase discouraging critical thought and meaningful discussion. This strategy dismisses or redirects a debate without addressing the heart of the argument, effectively shutting down further examination [44].
Example: "We shouldn't worry about the ethics of this; at the end of the day, it is what it is."

Vagueness (VA) Ambiguous words are shifted in meaning or left vague, potentially subject to skewed interpretations. This allows an arguer to be "vague enough to be right" while avoiding specific details that could be proven wrong [32]. It also includes *Equivocation* (i.e., changing the meaning of a term within an argument) [35, 36].
Example: "We will take appropriate measures to address the issue soon."

Chapter 2

Technical Background

This chapter introduces the technical foundations underlying the system presented in this work. Section 2.1 traces the history of Natural Language Processing (NLP), from early rule-based approaches to the neural methods that dominate the field today. Section 2.2 describes the Transformer architecture [48], the backbone of all modern NLP systems. Section 2.3 presents the encoder-only models employed in this work, starting from the foundational BERT model [49] and describing the specific variants used in our work: ALBERTo [50] and mmBERT [51]. Finally, Section 2.4 provides a brief overview of attribution methods, used to identify which parts of an input are most responsible for a model's prediction.

2.1 Evolution of NLP

Natural language processing (NLP) is a field of computer science and linguistics that focuses on using computational methods to process, understand, and generate human language. Because natural language is inherently ambiguous, context-dependent, and culturally loaded, building systems that process it reliably is an ongoing and complex challenge [52].

The history of NLP can be roughly divided into four eras. The first, spanning the 1950s through the 1980s, was dominated by *rule-based* approaches: linguists hand-crafted grammars, lexicons, and pattern-matching rules to parse sentences or translate text. While interpretable and precise in narrow domains, these systems were fragile and required significant expert effort to build and maintain, scaling poorly to new domains or languages.

The second era, from the late 1980s through the 2000s, shifted toward *statistical* approach. Rather than encoding linguistic knowledge manually, researchers trained probabilistic models on large corpora, exploiting the growing availability of digitized text and machine learning (ML) methods. Techniques such as *Naive Bayes*, *Hidden Markov Models* (HMMs), and *Support Vector Machines* (SVMs) achieved strong results on tasks like part-of-speech tagging, named entity recognition, and sentiment analysis [53]. The key insight was that useful linguistic regularities could be learned from data, rather than prescribed by hand.

The third era, emerging in the 2010s, was driven by the convergence of larger datasets, increased computational power, and advances in *neural* methods. The introduction of *dense word embeddings*, most notably Word2Vec [54] and GloVe [55], marked an important turning point: words were mapped to continuous vector spaces where semantic relationships were preserved, offering the first practical form of transfer learning in NLP. Building on these representations, recurrent architectures such as *Recurrent Neural Networks* (RNNs) and *Long Short Term Memory* (LSTMs) [56] could capture sequential dependencies across words, replacing most hand-crafted feature engineering.

However, recurrent models struggled with long-range dependencies and could not be efficiently parallelized. The introduction of the *attention* mechanism [57] and subsequently the *Transformer* architecture [48] in 2017 addressed these limitations directly, marking the beginning of the present era. Transformer-based models can capture long-range dependencies far more effectively and can be scaled to billions of parameters, becoming the backbone of modern *Large Language Models* (LLMs) and virtually every modern NLP system at the state of the art.

2.2 The Transformer

The Transformer [48] is a neural network architecture based on the attention mechanism. Its success stems from three key properties: (i) the ability to model long-range dependencies; (ii) full parallelization during training; and (iii) scalability: larger models trained on more data consistently yield better performance [58].

Self Attention Before entering the network, raw text is first split into subword units called *tokens*, and each token is mapped to its *embedding*. At this stage, embeddings carry only static, context-free meaning: the word *"bank"* always maps to the same vector, regardless of whether it appears near *"river"* or *"money"*. The core job of the transformer layer is to enrich these static representations with contextual information from the surrounding tokens.

This is achieved through *self-attention*. Intuitively, when processing a token, the model should be able to look at all other tokens in the sequence and decide which ones are most relevant. In the sentence *"The bank by the river was flooded"*, understanding *"bank"* requires attending to *"river"* and *"flooded"*. Self-attention makes this possible by allowing each token to gather information from the entire sequence in the context window and produce a new, context-enriched representation.

Formally, given a sequence of input tokens (X), each token embedding is projected into three distinct vectors through learnable weight matrices (W_Q, W_K, W_V), namely a *query* $(\mathbf{Q} = XW_Q)$, a *key* $(\mathbf{K} = XW_K)$, and a *value* $(\mathbf{V} = XW_V)$. The query represents what the current token is looking for, the key represents what each token has to offer, and the value carries the actual content to be aggregated. The attention output for each token is a weighted sum of all value vectors, where the weights are determined by the similarity (dot product) between that token's query and every other token's key:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{2.1}$$

Where d_k is the dimension of the key vectors, and the scores are scaled by $\sqrt{d_k}$ for numerical stability, and normalized via softmax so that they sum to one. The result is a new representation for each token that blends information from all positions in the sequence, weighted by relevance.

This mechanism is applied multiple times in parallel through *multi-head attention*. Each head maintains its own set of projection matrices W_i^Q, W_i^K, W_i^V, capturing different patterns from the input independently. One head might capture syntactic dependencies such as subject-verb agreement, while another captures semantic relationships. [59]

Architecture As illustrated in Figure 2.1 the transformer architecture follows a *encoder-decoder* structure. The encoder reads the full input and produces contextual representations, while the decoder generates the output sequence one token at a time, attending to both its own previous outputs and the encoder's representations.

Contrary to what the original paper suggests [48], *"attention is not quite all you need"*: The full Transformer is built by stacking multiple layers, each consisting of a multi-head self-attention sublayer followed by a *feed-forward network*, which applies a small MLP independently to each embedding to further encode the representations. *Residual connections* [60] and *layer normalization* [61] are applied around each sublayer to ensure stable training across deep stacks. Furthermore, since self-attention is *permutation-invariant* (i.e., it has no built-in notion of word order), positional information needs to be injected via *positional encodings* added to the token embeddings before the first layer.

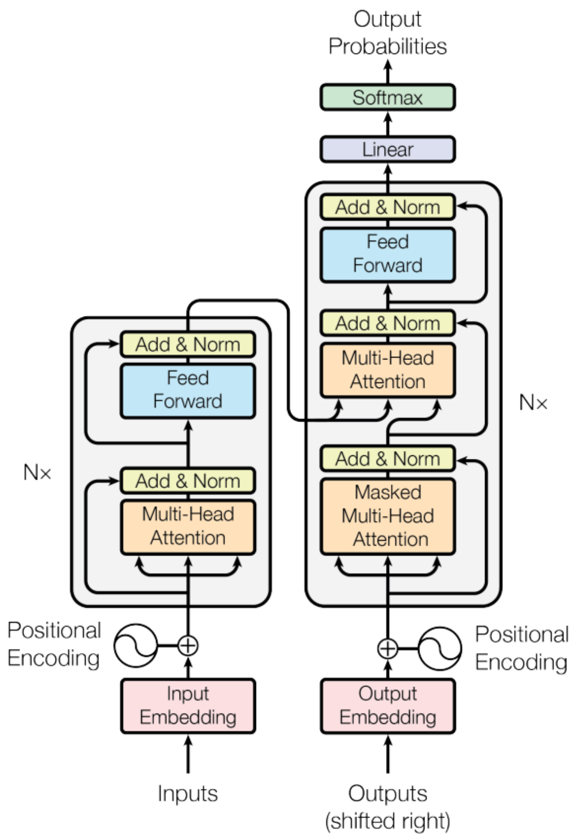


Figure 2.1: The original Transformer architecture [48], consisting of a stack of encoder layers (left) and decoder layers (right). Each layer combines multi-head self-attention with a position-wise feed-forward network, residual connections, and layer normalization. The decoder additionally incorporates cross-attention over the encoder's output.

Pre-training and Fine-tuning A further strength of the Transformer is how it supports *transfer learning*. Rather than training a model from scratch for each downstream task, a large Transformer is first *pre-trained* on vast amounts of unlabelled text to acquire general linguistic knowledge. This *Foundational Models* is then *fine-tuned* on a smaller labeled dataset for each specific target task. This two-stage approach dramatically reduces the amount of task-specific data required and consistently outperforms task-specific models trained from scratch.

Applications Although the original architecture was designed with an encoder–decoder structure, later work showed that each component can also be used independently depending on the task.

- *Encoder–decoder* models, such as T5 [62], retain the full architecture and are commonly used for tasks that require both understanding and generation, including machine translation and summarization.

- *Decoder-only* models, such as GPT [63], generate text autoregressively and form the foundation of most modern Large Language Models (LLMs). With appropriate prompting, their generative capabilities can be applied to a wide variety of downstream tasks.

- *Encoder-only* models, such as BERT [49] (Section 2.3), produce rich contextual representations of input tokens and are particularly effective for classification and labeling tasks, including sentiment analysis, named entity recognition (NER), and sequence labeling.

2.3 Pre-trained Encoder Models

In our work, we addressed AFD using fine-tuned encoder-only models. In this section, we describe the foundational models employed, whereas the system-specific architecture is discussed in Chapter 4.

BERT BERT (Bidirectional Encoder Representations from Transformers) [49] is the foundational encoder-only model and one of the most influential works in modern NLP. Its key feature is *bidirectional pre-training*: unlike earlier language models that processed text left-to-right, BERT conditions each token's representation on all surrounding tokens simultaneously, yielding richer contextual embeddings.

BERT was released in two sizes: BERT-Base, with 12 layers, 12 attention heads, and 110M parameters, and BERT-Large, with 24 layers, 16 attention heads, and 340M parameters. Both variants were pre-trained on the BooksCorpus [64] and English Wikipedia, totaling approximately 3.3 billion words.

Pre-training is performed using two self-supervised objectives:

- *Masked Language Modeling* (MLM), a random subset of input tokens is replaced with a special `[MASK]` token, and the model is trained to recover them from context. This forces the model to build a deep bidirectional understanding of language, since predicting a masked token requires integrating information from both sides of the sequence.

- *Next Sentence Prediction* (NSP), trains the model to predict whether two text segments are consecutive in the original document, encouraging it to capture discourse-level coherence beyond individual sentences.

Fine-tuning BERT for downstream tasks is straightforward: a final linear-softmax layer is added to produce label outputs. For sentence classification, a special `[CLS]` token is added in front of the input sequence; its final-layer representation is treated as a summary of the entire sequence and passed to a classification head trained on labeled data. For sequence labeling tasks, the per-token representations from the final encoder layer are used directly in the classification layer. In both cases, the full model (encoder and classification head) is re-trained on task-specific labeled data, allowing the pre-trained representations to adapt to the target domain while preserving the general linguistic knowledge acquired during pre-training.

Multi-task Fine-tuning An extension of the fine-tuning paradigm is *multi-task learning*, where a single encoder is fine-tuned on multiple tasks simultaneously. Rather than attaching a single task head, multiple heads are trained in parallel, each corresponding to a different task or label, while sharing the same underlying encoder. The shared encoder is forced to produce representations that are simultaneously useful across all tasks, which often improves generalization, particularly when tasks are related, and training data is limited [65]. In our work, we experimented with different multi-task configurations to address classification for multiple labels at both the sentence and token level, as described in Chapter 4.

BERT variations Since the introduction of BERT, numerous variants have been proposed. RoBERTa [66] improved the pre-training recipe by removing the NSP objective and training on larger batches and more data. DistilBERT [67] applied knowledge distillation to produce a model with 60% of BERT's parameters while retaining 95% of its performance. ALBERT [68] reduced memory usage through cross-layer parameter sharing and factorized embedding matrices. ELECTRA [69] replaced MLM with a more sample-efficient *replaced token detection* objective, where the model must distinguish real tokens from plausible substitutions. DeBERTa [70] introduced disentangled attention, treating token content and positional encodings as separate components throughout the attention mechanism. Finally, multilingual variants such as XLM-RoBERTa [71] extended pre-training to over 100 languages simultaneously.

In our work, we experimented with different encoders. The two most relevant are ALBERTo and Multilingual Modern BERT (mmBERT), described below.

- **ALBERTo** [50] is the Italian version of the original BERT-Base model with 12 layers and 100M parameters. It was pre-trained only on MLM using 191 GB of Italian Twitter data from the TWITA corpus [72], a large collection of Italian tweets.

- **mmBERT** [51] is a recent multilingual encoder with 307M parameters and 22 layers, supporting more than 1,800 languages. It introduces three main innovations: (i) a multi-stage learning paradigm composed of pre-training, mid-training, and decay phases; (ii) masked language modeling with a dynamic masking rate decreasing from 30% to 5%, encouraging progressively more nuanced language representations; (iii) dynamically varying data size and number of languages at each phase. Additionally, model temperature is dynamically adjusted to promote uniform sampling across languages and reduce bias toward high-resource ones.

2.4 Feature Attribution

Interpretability and Explainability Neural networks, including Transformer-based models, are highly effective but inherently opaque. While their performance can be evaluated externally, they provide little insight into *why* they behave in a certain way: their predictions emerge from billions of learned parameters in ways that are not directly human-readable. This limitation has motivated a growing body of research on *interpretability* and *explainability*, aimed at opening this black box. Following Roscher et al. [73], interpretability refers to the ability to map an abstract concept from a model into a form that is understandable to humans, whereas explainability is a stronger notion that additionally requires contextual justification for a model's behavior. These properties are valuable for several reasons: they enable debugging and auditing of model behavior, help identify potential biases, and most importantly, allow researchers to gain scientific insight into the phenomena being modeled, rather than treating the model as a closed system.

Attribution In the context of NLP, the most common form of interpretability is *feature attribution*: given a model prediction, the goal is to assign a relevance score to each input token, indicating how much it contributed to that decision. Attribution methods can thus reveal which parts of a text a model relied on when producing a given output. In our work, we explored the use of attribution as a mechanism for span extraction: given a prediction at the post level, attribution scores are used to identify the exact text span triggering the prediction, as described in Section 4.4.

Attribution Methods Attribution methods for Transformer-based models can be broadly grouped into three families [74].

- **Perturbation-based methods** (e.g., SHAP [75], LIME [76]) measure how the model's output changes as input features are repeatedly masked or altered. While model-agnostic and conceptually simple, they are computationally expensive, requiring many forward passes, and often fail to explain hidden representations

- **Attention-based methods** (e.g., Attention Rollout [77]) use the internal attention weights of the model as a proxy for token importance. These methods are computationally efficient and straightforward to implement, but they provide explanations only for the softmax output within attention modules, ignoring other critical components such as Feed-Forward Networks, where recent work has shown that factual knowledge is actually stored [78, 79]. Moreover, attention-based methods are not class-specific and suffer from low resolution in the resulting attribution maps, producing noise and undesirable artifacts [74].

- **Backpropagation-based methods** trace relevance backwards through the network, from the output to the input. The simplest variant, *Input* $\times$ *Gradient* [80], linearizes the model using the gradient of the prediction with respect to the input embeddings as a proxy for importance. While efficient, this approach is prone to noisy attributions due to the *gradient shattering* effect in deep networks [81]. *Layer-wise Relevance Propagation* (LRP) [82] addresses this by decomposing individual layer functions instead of linearizing the entire model. It uses customized propagation rules for each layer type, designed to maintain a *conservation property*: the total relevance is preserved as it is distributed backwards through the network. This ensures more faithful attributions while retaining computational efficiency.

AttnLRP In our work, we used *AttnLRP* (Attention-Aware Layer-Wise Relevance Propagation) [74], a propagation-based method that extends LRP specifically for Transformer architectures. Indeed, a key challenge in applying standard LRP to Transformers is the attention mechanism, which involves several operations for which standard LRP rules do not straightforwardly apply: the softmax non-linearity, the bi-linear query-key matrix multiplication, and normalization layers. AttnLRP addresses each of these by deriving novel, mathematically grounded propagation rules within the Deep Taylor Decomposition framework [83]. The result is a token-level relevance map indicating how positively or negatively each input token contributed to a given prediction. As demonstrated in Achtibat et al. [74], AttnLRP stands out for its unique combination of simplicity, faithfulness, and efficiency: it consistently outperforms attention-based and other backpropagation-based approaches in terms of faithfulness across multiple model architectures, while requiring only a single backward pass, yielding $\mathcal{O}(1)$ computational complexity comparable to standard gradient methods.

Chapter 3

The FadeIT Shared Task

This chapter describes FadeIT [2], the shared task that motivates and frames the work presented in this thesis. Section 3.1 introduces the task and its two subtasks. Section 3.2 discusses the perspectivist approach to annotation and motivates its adoption for the inherently subjective task of fallacy annotation. Section 3.3 presents the FAINA dataset [4], describing its data collection procedure, annotation process, and key statistics. Section 3.4 details the data formats used for the two subtasks. Finally, Section 3.5 describes the evaluation metric and provides a brief overview of the baseline systems and other teams' participations.

3.1 Task Description

FadeIT (**Fa**llacy **de**tection in **IT**alian Social Media Texts) [2] is a shared task organized within the **EVALITA 2026** campaign[1], a periodic initiative for the evaluation of NLP and speech technologies for Italian, promoted by the Italian Association for Computational Linguistics (AILC), and held in Bari in February 2026 [3].

FadeIT is the first shared task on Automatic Fallacy Detection (AFD) for Italian social media text and is based on the FAINA dataset [4], described in Section 3.3.

The task is organized into two subtasks of increasing complexity:

- **Subtask A: Coarse-grained fallacy detection.** Given the text of a social media post, the participants are asked to predict all fallacies expressed in it. This is a *multi-label classification* task over 20 fallacy classes: a post can be assigned zero or more labels simultaneously.

- **Subtask B: Fine-grained fallacy detection.** Given the text of a social media post, the participants are asked to identify the exact text spans corresponding to each fallacy expressed in it; different fallacies may overlap, partially or in full, with each other. This is a *multi-label sequence labeling* task, requiring both span localization and fallacy classification.

The two subtasks are related but distinct: Subtask A requires understanding *which* fallacies are present in a post, while Subtask B requires identifying *where* exactly they occur in the text. As discussed in Chapter 4, this relationship has been exploited through multi-task learning.

[1]https://www.evalita.it/campaigns/evalita-2026/

3.2 The Perspectivist Approach

A core design choice underlying both FAINA and FadeIT is the adoption of a *perspectivist* approach to annotation. In classical NLP dataset construction, annotations are collected from experts or crowd workers following a predefined annotation scheme. Once a sufficient number of instances have been labeled, disagreements between annotators are resolved to produce a single *gold standard*, for example through majority voting, adjudication by an expert, or averaging numeric values. This practice, known as the *prescriptive* paradigm [84], treats disagreement as a problem to be eliminated, under the unwritten assumption that *there is exactly one ground truth*.

This approach is well-suited for tasks involving objective properties, often technical linguistic aspects of texts such as part-of-speech tagging or syntactic parsing, where a single truth is a reasonable assumption. However, it has been recently challenged for inherently subjective tasks where disagreement should not be considered as noise, but *signal* [85].

Plank [85] frames this as the *problem of human label variation*: when a task involves subjective interpretation, multiple annotators can each be correct while assigning different labels, and collapsing their judgments into a single answer discards valuable information about the genuine ambiguity of the phenomenon being modeled. This concern is particularly relevant for tasks such as sentiment analysis or hate speech detection, where annotators' cultural, ethnic, and social backgrounds can have a significant impact on their judgments. Related work further emphasizes the role of subjectivity [86], the validity of multiple plausible answers [87], and the idea that annotator disagreement reflects meaningful human perspectives rather than annotation errors [88].

The *Perspectivist* manifesto[2] formalizes this position for the NLP community, arguing that models should be trained and evaluated to account for the full distribution of human perspectives rather than a single aggregated label [89]. In this view, a dataset that preserves multiple, individually valid annotations is a *richer* and more honest representation of the underlying phenomenon.

Subjectivity of Fallacy Annotation Fallacy annotation is a remarkable case for perspectivism as it is inherently subjective. As discussed in Chapter 1, there is no universally accepted definition or taxonomy of fallacies, and their identification often requires interpreting different subjective factors, such as: the speaker's intent, the audience's background, and the broader context.

Consider the phrase *"Are you with America? Vote YES!"*. One annotator may label it as *Flag Waving*, reading it as an appeal to national identity. Another may additionally identify a *Slogan*, given its brevity and rhetorical force. A third may infer the implicit claim *"or you must be against America"*, identifying a *False Dilemma*. All three readings are coherent and defensible given the definitions, yet they produce diverging annotations.

Annotators may also disagree on whether a label applies at all, depending on their individual thresholds for what constitutes a fallacy. Consider the phrase *"migrants are flooding our country"*. One annotator may flag *migrants* as *Name Calling* or interpret the flooding metaphor as *Loaded Language* and/or *Appeal to Emotion*, reading the phrasing as rhetorically charged. Another annotator may treat it as a

[2] `https://pdai.info`

neutral, technically accurate description of a demographic phenomenon, assigning no label. Neither reading is wrong: the question of where ordinary descriptive language ends and fallacious framing begins is genuinely ambiguous.

Different degrees of familiarity with the topic of the arguments can also play an important role. A sentence such as *"Use disinfectants or you will get Covid-19!"* may be read as a plausible factual claim by an annotator with no epidemiological expertise; however, an annotator who knows that Covid-19 does not spread primarily via contaminated surfaces could read it as an exaggerated and misleading warning, potentially labeling it as *Causal Oversimplification*, and *Appeal to Emotion*.

Background knowledge is particularly important in the context of social media, where brief texts rely heavily on shared contextual knowledge, cultural references, and rapidly evolving semantic conventions, including *algospeak*, the practice of substituting words with coded alternatives to evade content moderation algorithms [90]. A comment such as *"it is always like that 🧃"* may appear innocent at first glance; however, the juice emoji (🧃) has been used as a coded derogatory reference to Jewish people, meaning that an annotator aware of this convention would recognize the phrase as a potential instance of *Name Calling or Labelling*, while another would see no fallacy at all. The subjectivity is not about personal degree, but a complete knowledge asymmetry. The two annotators are, in a meaningful sense, reading different texts.

Finally, span-level annotation further increases the complexity of the problem, as annotators may agree on the presence of a fallacy but disagree on the exact sequence of tokens that best represents it in the text.

For these reasons, and in line with the perspectivist approach, two expert annotators independently labeled the FAINA dataset, producing two equally valid annotations that were not reconciled into a single ground truth as described in Section 3.3.2. This decision propagates into the FadeIT evaluation framework, where systems are assessed against both annotations simultaneously, as detailed in Section 3.5.1

3.3 The FAINA Dataset

FAINA (**FA**llacy detection with **IN**dividual **A**nnotations) is introduced by Ramponi et al. [4] as the first dataset for fallacy detection following the perspectivist approach. It consists of 1,440 Italian posts about topics such as migration, climate change, and public health, covering a four-year time window and annotated at the span level by two expert annotators using the 20 fallacy classes.

Figure 3.1 illustrates an example from FAINA of two different, but equally valid, annotations for the same post, provided by the two expert annotators.

Fallacy taxonomy The 20 fallacy types in FAINA were derived by reviewing and harmonizing definitions from prior work, starting from a candidate pool of 41 fallacies and consolidating those with similar definitions [44, 32]. Furthermore, the 20 types are organized into a taxonomy grouping them into three macro-categories: *Insufficient Proof*, *Simplification*, and *Distraction* (Figure 1.1). The full list of fallacy classes is described in detail in Section 1.5 and summarized in Appendix A.

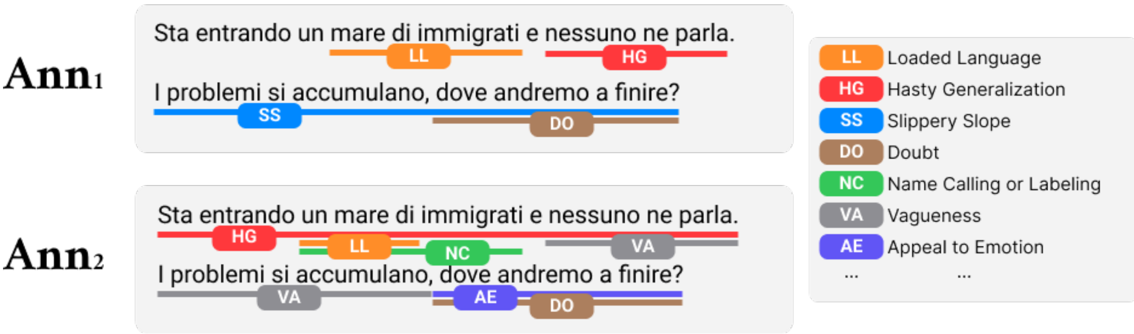


Figure 3.1: Example of different annotations provided by the two annotators (en: *"A flood of immigrants is coming in and nobody is talking about it. Problems keep piling up, where are we going to end up?"*)

3.3.1 Data Collection

Posts were collected from Twitter/X using the platform's APIs[3], targeting Italian-language content on three socially relevant topics: *migration*, *climate change*, and *public health*. These domains were selected because they are often characterized by a high presence of fallacious argumentation, as well as their strong societal impact, particularly in the context of online misinformation and public discourse [32, 29]. Furthermore, to minimize temporal and topical biases, posts were collected over a four-year window from January 2019 to December 2022

Candidate posts were filtered using a manually curated list of 436 domain-specific keywords, and only posts with at least five tokens were retained to ensure sufficient argumentative content. Among the filtered posts, those with the highest number of likes and retweets were selected, focusing on high-impact messages [91]. To further prevent topic, author, and temporal bias, the final dataset was balanced by selecting the top-k posts ($k = 10$) for each month-topic combination (e.g., top 10 posts about "migration" posted in 2021-01), excluding multiple posts from the same author.

The result is a dataset of 1,440 posts, balanced across topics (480 per topic) and time (360 per year), for a total of 58,490 tokens.

3.3.2 Annotation Procedure

Annotation was performed by two expert annotators (Ann1 and Ann2) following a prescriptive protocol [84] that combines structured guidelines with multiple discussion rounds. The task was considered too complex and nuanced for crowd sourcing, as also argued by Da San Martino et al. [44] for the related task of propaganda detection. Annotation required annotators to identify all text segments containing fallacies and assign each one or more of the 20 labels from the FAINA inventory. The process took approximately 380 person-hours in total, including discussions.

The annotation was conducted in five rounds of increasing size. At each round, annotators independently labeled all posts, then discussed those that diverged in the assigned fallacy type, span extent, or both. This phase minimizes disagreements due to errors or attention drops, resolving them while explicitly preserving naturally occurring variation and genuine disagreement. After all rounds, both annotators performed a final revision to ensure cross-round consistency.

[3] Data was retrieved in February 2023, when APIs for research purposes were still freely available.

Inter-annotator agreement Inter-annotator agreement (IAA) for FAINA was measured using γ [92] and γ_{cat} [93], implemented via the `pygamma-agreement` package. Unlike classical token-level coefficients such as Krippendorff's α, these metrics are better suited for span-level annotation with variable-length, potentially overlapping segments: γ quantifies the degree of agreement in *span identification*, while γ_{cat} additionally accounts for the correctness of *span classification*. The final IAA on the complete dataset is $\gamma = 0.624$ for span identification and $\gamma_{cat} = 0.544$ for span classification [4], comparable to results reported for the related task of propaganda detection [44]. Notably, the lower γ_{cat} relative to γ suggests that annotators agree more on span boundaries than on fallacy type. Crucially, analysis across annotation rounds reveals that the upper bound of achievable agreement remains stable ($[0.6, 0.7]$ for γ and $[0.5, 0.6]$ for γ_{cat}) regardless of how many discussion rounds were conducted. This plateau confirms that human label variation can be resolved only partially and that the residual disagreement reflects genuine interpretive variability intrinsic to fallacy annotation, as previously noted in Section 3.2.

3.3.3 Dataset Statistics

Overall, FAINA consists of 11,064 annotated spans ($5,532 \pm 253$ per annotator) across 1,440 posts and 58,490 tokens. Table 3.1 reports the number of spans, average span length (in tokens), and per-class IAA (γ_{cat}) for each fallacy type.

Table 3.1: Number of spans, average span length (tokens), and IAA (γ_{cat}) per fallacy type in the FAINA dataset [4].

Id	**Fallacy type**	**Spans**	**Length**	γ_{cat}
AH	Ad hominem	319	16.0 ± 13.3	0.6651
AA	Appeal to authority	213	6.4 ± 4.3	0.6147
AE	Appeal to emotion	2,049	5.1 ± 4.9	0.4730
CO	Causal oversimplification	142	19.0 ± 10.7	0.5282
CP	Cherry picking	94	28.8 ± 12.3	0.3415
CR	Circular reasoning	20	26.8 ± 11.0	0.5397
DO	Doubt	482	16.1 ± 11.4	0.7103
EP	Evading the burden of proof	406	16.2 ± 9.9	0.4335
FA	False analogy	239	22.1 ± 13.4	0.5243
FD	False dilemma	90	15.9 ± 11.0	0.5568
FW	Flag waving	393	4.3 ± 4.9	0.5735
HG	Hasty generalization	464	11.2 ± 8.0	0.4980
LL	Loaded language	2,484	2.5 ± 2.7	0.4365
NC	Name calling or labeling	1,124	2.6 ± 1.7	0.5566
RH	Red herring	257	13.0 ± 8.5	0.4378
SS	Slippery slope	172	10.8 ± 6.8	0.6552
SL	Slogan	384	3.5 ± 3.1	0.7101
ST	Strawman	109	36.3 ± 15.4	0.5570
TC	Thought-terminating cliché	285	5.2 ± 3.0	0.5305
VA	Vagueness	1,338	9.1 ± 8.6	0.3701
	All	**11,064**	$\mathbf{7.6 \pm 9.3}$	**0.5445**

Span Length Fallacy spans have an average length of 7.6 ± 9.3 tokens, but length varies considerably across classes. The shortest fallacies are those related to language use [8], such as *Loaded Language*, *Name Calling or Labeling*, and *Slogan* (<4 tokens on average), while the longest are those commonly referred to as logical fallacies and fallacies of diversion [8], such as *Strawman*, *Cherry Picking*, and *Circular Reasoning* (>25 tokens on average). This distinction has a direct impact on detection performance, as discussed in Chapter 4: shorter, lexical fallacies tend to rely on identifiable surface cues and are easier to detect, while longer, logical fallacies require broader contextual understanding and are considerably more challenging

Span Overlap The dataset is densely annotated (3.8 ± 0.2 spans per post), and overlaps among fallacy spans are very frequent. As shown in Figure 3.2, for most fallacy types, at least half of the annotated tokens co-occur with tokens from another fallacy. This density of overlapping, multi-label annotations at the span level distinguishes FAINA from all prior fallacy detection datasets, which either provide coarse post-level labels or assume non-overlapping spans.

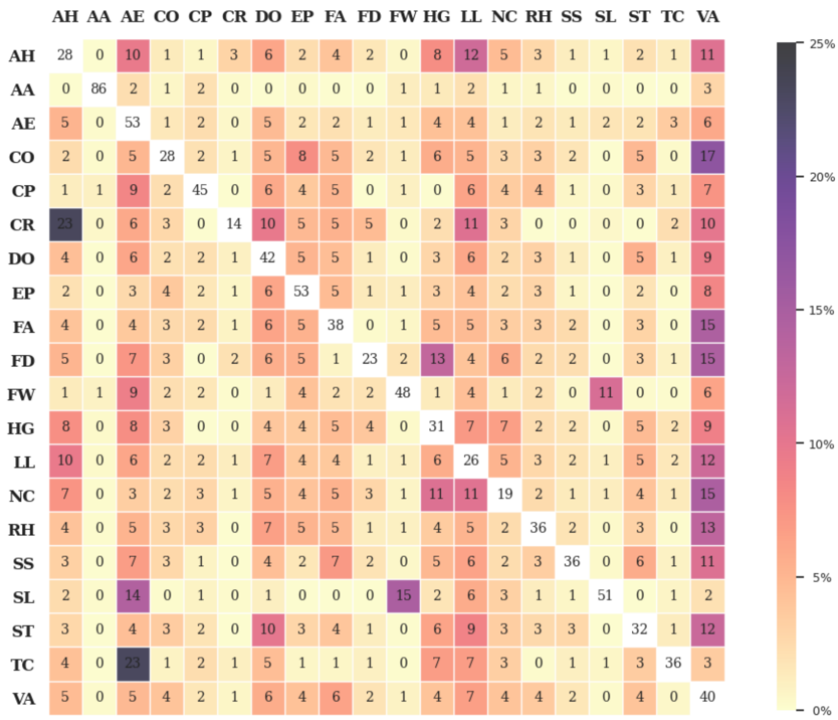


Figure 3.2: Overlap of fallacy annotations in terms of token percentages. Each row indicates the percentage of tokens for a given fallacy type that overlaps with any other fallacy type (columns). White cells (diagonal) indicate the percentage of tokens for each fallacy type that do not overlap with any other fallacy type [4].

Train/test split For the FadeIT shared task, participants were provided with 80% of the dataset (1,152 posts) for training and validation, while the remaining 20% (288 posts) was held out as a test set. Among the released portion, there are 4,124 unique fallacy annotations, counting a fallacy once per post when assigned by both annotators. corresponding to an average of 3.58 fallacies per post.

Table 3.2: Number and percentage of posts annotated with each fallacy type in the released portion of FAINA (1,152 posts) [4]. Posts may contain multiple fallacy labels; when a fallacy is assigned by both annotators to the same post, it is counted once.

Id	**Fallacy**	**Count**	**Percentage (%)**
LL	Loaded language	699	60.68
AE	Appeal to emotion	652	56.60
VA	Vagueness	534	46.35
NC	Name calling or labeling	362	31.42
HG	Hasty generalization	225	19.53
DO	Doubt	214	18.58
EP	Evading the burden of proof	204	17.71
FW	Flag waving	158	13.72
AH	Ad hominem	140	12.15
TC	Thought-terminating cliché	138	11.98
SL	Slogan	130	11.28
RH	Red herring	129	11.20
FA	False analogy	125	10.85
AA	Appeal to authority	97	8.42
SS	Slippery slope	84	7.29
CO	Causal oversimplification	68	5.90
ST	Strawman	58	5.03
CP	Cherry picking	58	5.03
FD	False dilemma	39	3.39
CR	Circular reasoning	10	0.87

Class distribution To enable a post-level version of the task (Subtask A), span-level annotations were also transposed to the post level. Table 3.2 summarizes the number and percentage of posts containing each fallacy in the released portion of the dataset. The distribution is highly skewed: a small number of classes dominate, with *Loaded Language* (699 posts, 60.7%), *Appeal to Emotion* (652, 56.6%), and *Vagueness* (534, 46.4%) appearing in the majority of posts, while several classes are extremely rare, for instance *Circular Reasoning* appears in only 10 posts (0.87%). This imbalance poses a significant challenge for the classification task, as discussed in Chapter 4.

3.4 Data Format

The FAINA dataset is distributed in two different formats depending on the subtask.

Subtask A For post-level fallacy detection, the data is provided in a tab-separated format with a header line. Each line encodes information about a single post; namely, its identifier, date, topic, text, and fallacy labels. Post-level annotations by each of the two annotators are provided in separate columns; when the same annotator assigns multiple labels to a post, these are separated by a pipe character (|) and ordered lexicographically. When no label is assigned by an annotator, the corresponding column is left empty.

Subtask B For span-level fallacy detection, the data follows a format inspired by the CoNLL format. Posts are separated by blank lines. Each post begins with a header block providing post-level metadata (id, date, topic keywords, and text), followed by one line per token. Token-level information is tab-separated. Annotations follow the BIO tagging scheme (B: begin, I: inside, O: outside), and multiple labels for the same token and annotator are separated by a pipe. Each label follows the format `$BIO-$LABEL` (e.g., `B-Vagueness`, `I-Strawman`), ordered lexicographically by `$LABEL` in the multi-label case.

Both formats are illustrated in Table 3.3. The variables are defined as follows:

- `$POST_ID`: the anonymized identifier of the post;
- `$POST_DATE`: the date of the post in `YYYY-MM` format;
- `$POST_TOPIC`: the topic of the post (*migration*, *climate change*, or *public health*);
- `$POST_TEXT`: the anonymized[4] text of the post;
- `$LABELS_BY_ANN_j`: the fallacy label(s) assigned by annotator j at the post level;
- `$TOKEN_i`: the index of the i-th token within the post;
- `$TOKEN_i_TEXT`: the text of the i-th token;
- `$TOKEN_i_LABELS_BY_ANN_j`: the BIO-tagged fallacy label(s) assigned by annotator j for the i-th token.

Table 3.3: Data format for subtask A (*top*) and subtask B (*bottom*)

```
$POST_ID  $POST_DATE     $POST_TOPIC  $POST_TEXT   $LABELS_BY_ANN_1  $LABELS_BY_ANN_2
```

```
# post_id = $POST_ID
# post_date = $POST_DATE
# post_topic_keywords = $POST_TOPIC
# post_text = $POST_TEXT
$TOKEN_1  $TOKEN_1_TEXT  $TOKEN_1_LABELS_BY_ANN_1  $TOKEN_1_LABELS_BY_ANN_2
…
$TOKEN_N  $TOKEN_N_TEXT  $TOKEN_N_LABELS_BY_ANN_1  $TOKEN_N_LABELS_BY_ANN_2
```

3.5 Participation

3.5.1 Evaluation

Systems are evaluated coherently with the perspectivist nature of the dataset (Section 3.2): rather than measuring predictions against a single ground truth, performance is assessed against the two equally valid gold standards, one per annotator, averaged over the held-out 20% test set.

[4]User mentions, URLs, email addresses, and phone numbers are replaced with [USER], [URL], [EMAIL], and [PHONE] placeholders, respectively.

For Subtask A, evaluation relies on standard *micro-averaged* precision, recall, and F1 score. Systems are ranked by micro F1 score.

For Subtask B, evaluation uses *micro-averaged* precision, recall, and F1 score, designed for span-level annotations with potential overlaps, introduced by Da San Martino et al. [44] and extended to operate at the token level, so that partial credit is proportionally assigned to partial span matches based on token overlap.

Furthermore, two evaluation modes are used: *strict*, where a prediction is rewarded only if it matches the intended fallacy label exactly, and *soft*, in which partial credit is awarded when the predicted label is an immediate parent of the gold label in the fallacy taxonomy. For instance, predicting *Red herring* instead of *Appeal to authority* is less problematic than predicting *False dilemma*. Systems are ranked by micro F1 score under soft evaluation.

3.5.2 Baselines

The shared task organizers provided two families of encoder-based multi-task baseline systems, both introduced in the original FAINA paper [4]. Both families are available in two variants, depending on the backbone encoder: ALBERTo [50] or UmBERTo [94], yielding MVML-ALB/MVML-UMB and MVMD-ALB/MVMD-UMB respectively. Among the two variants, the UmBERTo-based variants were adopted as the official baselines for the shared task, as they represent an easier starting point to beat on this difficult task.

- **MVML (Multi-View Multi-Label)** Used as baseline for subtask A, it fine-tunes a shared encoder jointly on two binary multi-label classification tasks, one per annotator, and outputs all fallacy labels whose predicted score exceeds a fixed threshold.

- **MVMD (Multi-View Multi-Decoder)** Used as baseline for subtask B, it frames span detection as a set of BIO sequence labeling tasks, for each combination of fallacy class and annotator, for a total of 40 tasks.

3.5.3 Participating Teams

FadeIT attracted 25 runs by 7 teams from five countries, making it one of the most participated shared tasks at EVALITA 2026 [3]. For Subtask A, 16 runs were submitted by 6 teams; for Subtask B, 9 runs by 3 teams. The participating teams and their core approaches are summarized below; for full details and ranking, we refer to the shared task overview [2] and the respective system description papers.

- **Kenji-Endo** [95] (Subtask A) The team trained a decoder-only causal language model from scratch on different Italian corpora to address multiple EVALITA tasks; for FadeIT, the model was fine-tuned on FAINA data in a discriminative setting, accounting for class imbalance in the loss function.

- **Label** [96] (Subtask A) The team proposed a two-step LLM prompting approach: a first step to produce a fallacy-specific textual analysis for each of the 20 fallacy types independently, and a second step to assign a numeric score indicating the degree of fallacy presence. These scores are combined with topic metadata and aggregate statistics as features for a supervised MLP trained to predict each annotator's labels separately.

- **MALTO** [97] (Subtask A) The team fine-tuned an encoder-based model for multi-label classification, using a global threshold and experimenting with different loss configurations across runs to address class imbalance.
- **PuDy** [98] (Subtask B) The team adopted a span-based modeling approach, representing candidate spans as concatenations of boundary embeddings and a learned span length embedding. Their most distinctive contribution is a hierarchy-aware label propagation strategy that provides training supervision not only to the gold fallacy type but also to its parent in the FAINA taxonomy, explicitly leveraging the taxonomic structure of the fallacy inventory.
- **RBG-AI** [99] (Subtasks A and B) The team applied a unified few-shot prompting framework using instruction-tuned LLMs, selecting examples to maximize diversity across fallacy types and annotation views. To manage the large output space, predictions are constrained to coarse-grained fallacy groups derived from semantic closeness and corpus-level distributional statistics.
- **UNICA** [100] (Subtask A) The team built a retrieval-augmented generation (RAG) pipeline in which each post is converted to embeddings and the most semantically similar training examples are dynamically retrieved from a vector database to construct a few-shot prompt for an LLM.

Our team,**TiGRO** [1], participated in both subtasks with three runs each, experimenting with multi-task learning configurations over fine-tuned encoder-based models. The system design, experimental setup, and discussion are described in detail in Chapter 4.

Chapter 4

TiGRO : *E Pluribus Unum*

This chapter presents the systems developed by the **TiGRO** team for the FadeIT shared task. Section 4.1 describes the system architecture, including the choice of encoders, the handling of the two annotators, the modeling approaches, and the training setup. Section 4.2 reports the results on both subtasks, with per-class breakdowns and comparisons against an SVM baseline. Section 4.3 analyzes these results in depth, discussing the influence of class frequency, span length, and lexical surface cues on model performance. Finally, Section 4.4 discusses alternative approaches explored during development but ultimately not submitted, including data integration strategies and attribution-based span extraction.

Team Overview The TiGRO (Turin in GROningen) team participated in the FadeIT shared task as a joint effort between the University of Turin and the University of Groningen, made possible by the Erasmus+ program. We submitted three systems for Subtask A, targeting post-level fallacy detection, and three systems for Subtask B, targeting span-level fallacy identification. We experimented with fine-tuning encoder-based models across different multi-task learning configurations. Our best submitted system ranked **first** for Subtask A among sixteen submissions (micro-F1: **56.39**). For Subtask B, it ranked fourth among nine submissions under soft evaluation (micro-F1: **44.98**), and first under strict evaluation (micro-F1: **42.13**).

4.1 System Description

Our general strategy was to identify learning approaches that could maximize performance on both subtasks simultaneously. Although the two subtasks are evaluated independently in the shared task, they are deeply related: a fallacious argument manifests at the post level as a label and at the span level as a localized textual trigger. Knowing that a specific fallacy is present in a post can help identify the corresponding span, and vice versa. This motivated our central design choice to explore joint modelling of both subtasks, also inspired by the multi-task encoder baselines introduced in the original FAINA paper [4] and described in Section 3.5.2.

In all our systems, we opted to fine-tune encoder-based models rather than using in-context learning with generative LLMs. This decision was guided by two considerations: first, LLMs tend to underperform when prompted to predict across a large number of classes simultaneously [101, 37]; second, prior work on the FAINA dataset specifically showed that LLMs struggle with this task [4]. This choice proved well-founded, as reported in the shared task overview [2], encoder-based systems consistently outperformed LLM-based approaches across both subtasks.

Based on the results obtained on our internal development split, we selected six systems for official submission, three per subtask. Table 4.1 provides an overview of all submitted systems; the remainder of this section describes the modeling choices in detail.

Table 4.1: Overview of the submitted TiGRO models.

System	Approach	Encoder	Subtask
TiGRO-A1	Single-task (one vs rest)	mmBERT	A
TiGRO-A2	Multi-task (post only)	ALBERTo	A
TiGRO-A3	Multi-task (post and span)	ALBERTo	A
TiGRO-B1	Multi-task (span only)	ALBERTo	B
TiGRO-B2	Multi-task (post and span)	ALBERTo	B
TiGRO-B3	Multi-task (post and span)	mmBERT	B

Encoders For the encoder backbone, we experimented with several pretrained encoders. The two models used in our submissions are ALBERTo and mmBERT (Multilingual Modern BERT), described in Section 2.3. ALBERT was selected as an Italian-specialized encoder, also adopted in the FAINA baselines (Section 3.5.2), where it showed promising performance compared to UmBERTo. mmBERT was selected as a larger, more advanced multilingual model for comparison. Despite these advantages, ALBERTo often performed better in our experiments, likely due to its Italian specialization and Twitter pretraining, which closely match the FAINA domain. For each submitted system, we selected the encoder performing best on the development set.

Handling Two Annotators As described in Chapter 3, the FAINA dataset follows a perspectivist annotation philosophy: rather than resolving disagreements into a single ground truth, it retains and evaluates against both annotators' labels independently. We embrace this view, but instead of treating the two annotation sets as separate training signals, we merge them into a single unified annotation. Concretely, the label set assigned to each post or token is the *union* of the labels from both annotators, so that all perspectives are preserved and considered valid. At the span level, when two spans of the same fallacy class overlap but differ slightly in their boundaries, we retain a single span covering the union of both. Beyond the theoretical motivation, this choice was also practically driven by the limited size of the dataset: splitting the annotation signal across two annotators for 20 classes would substantially increase the training difficulty. Empirically, this merging strategy led to consistent performance improvements across all tested architectures.

Modeling Approaches One overall decision we made, independently of other implementation choices, is to address fallacy labeling (Subtask A) as a collection of binary tasks. Rather than dealing with a multi-class problem with 20 classes, we deal with 20 different binary tasks, one per fallacy. Subtask B is framed analogously as a set of 20 BIO sequence labeling tasks, one per fallacy class. This formulation simplifies each individual classification, while naturally accommodating the multi-label nature of the task, resulting in 20 independent binary decisions per post.

For Subtask A, we experimented with both single- and multi-task approaches. In the single-task setup, we fine-tuned 20 independent binary classifiers, one for each fallacy type (**TiGRO-A1**).

In multi-task learning, we experimented with leveraging interactions both across classes within Subtask A as well as between Subtask A and Subtask B. In one setting, we trained a multi-task classifier by fine-tuning 20 parallel tasks, which jointly learn binary fallacy classification at the post level for Subtask A (**TiGRO-A2**). For Subtask B, we adopted the same multi-task strategy but framed the problem as BIO tagging at the span level (**TiGRO-B1**). Finally, in three of our systems, we jointly address *both* the fallacy classification at the post level *and* the span detection for a total of 40 tasks (**TiGRO-A3**, **B2**, **B3**).

Our systems can be summarized as follows:

- **TiGRO-A1**: 20 binary post-level classifiers trained *independently*, each with its own encoder (Subtask A only).
- **TiGRO-A2**: 20 binary post-level classifiers trained *jointly* over a single shared encoder (Subtask A only).
- **TiGRO-B1**: 20 BIO sequence labeling tasks trained *jointly* over a single shared encoder (Subtask B only).
- **TiGRO-A3 / B2 / B3**: 20 post-level and 20 span-level tasks trained *jointly* over a shared encoder, for a total of 40 tasks (both Subtasks).

It is worth noting that, **TiGRO-A3** and **TiGRO-B2** are identical models, submitted separately to each subtask. **TiGRO-B3** adopts the same 40-task architecture but with a different encoder (mmBERT instead of ALBERTo). When Subtask A and Subtask B share the same encoder, no explicit dependency is imposed between the two subtasks.

Best Model Architecture The best-performing systems for both subtasks share the same architecture, illustrated in Figure 4.1: a single pre-trained encoder fine-tuned jointly on 20 post-level binary classification tasks and 20 span-level BIO tagging tasks, for a total of 40 task heads. Each task head consists of a linear layer applied either to the `[CLS]` token representation (for Subtask A) or to the per-token representations (for Subtask B). The only difference between the two best systems is the choice of encoder: ALBERTo for **TiGRO-A3** and mmBERT for **TiGRO-B3**, with the selection based on development set performance.

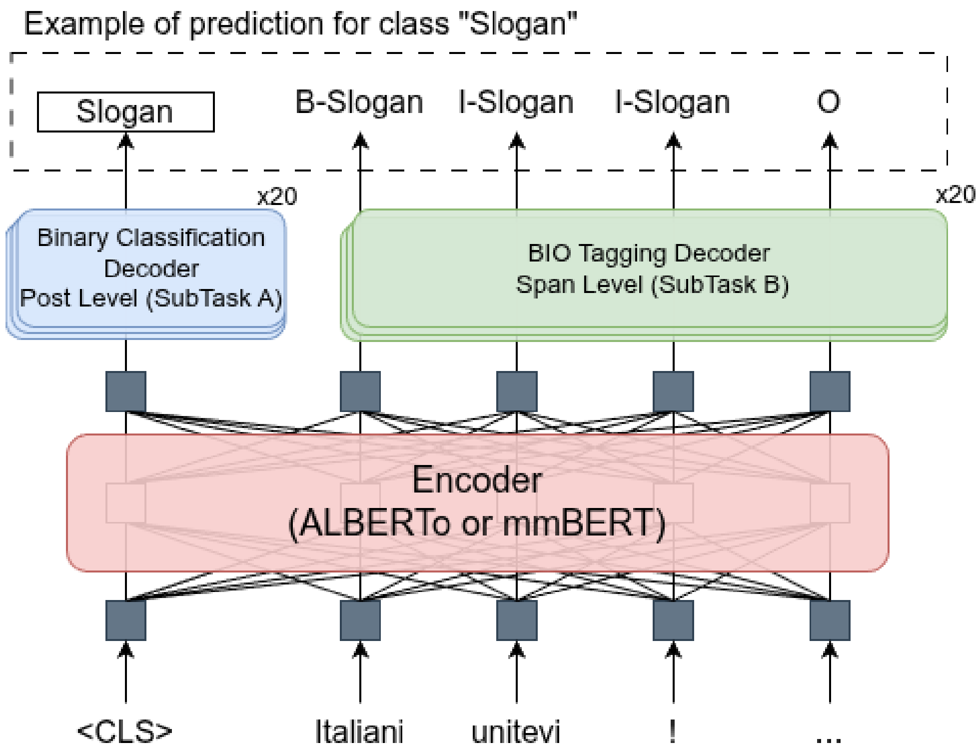


Figure 4.1: Architecture of the best-performing TiGRO models (**TiGRO-A3** / **TiGRO-B3**). A shared encoder is fine-tuned jointly over 20 post-level binary classification heads (Subtask A) and 20 BIO sequence labeling heads (Subtask B), for a total of 40 tasks.

Training Setup The train-dev data released by the shared task organizers (1,152 posts) was split into a training set (80%, 921 instances) and a development set (20%, 231 instances), corresponding respectively to 64% and 16% of the full FAINA dataset. The same fixed split was used across all experiments to ensure comparability during system development. For the final submissions, the best models were retrained on the full 1,152 instances.

All multi-task systems were trained using the MaChAmp v0.4 toolkit [102] with the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.99$), batch size of 32, learning rate of 1e-4, encoder dropout of 0.2, and for 20 epochs. Full hyperparameter details for each system are reported in Table 4.2. All experiments were run on the Hábrók high-performance computing cluster at the University of Groningen, using an NVIDIA A100 GPU.

Table 4.2: Training hyperparameters for all submitted TiGRO models.

Models	Encoder	Epochs	Batch size	LR	Dropout
TiGRO-A1	mmBERT	3	16	2e-5	0.0
TiGRO-A2	ALBERTo	10	32	1e-4	0.2
TiGRO-A3, B1, B2	ALBERTo	20	32	1e-4	0.2
TiGRO-B3	mmBERT	20	32	5e-4	0.2

4.2 Results

Overall, the TiGRO family turned out to be very competitive across both tasks. For Subtask A, **TiGRO-A3** achieved first place (#1) among fifteen submissions (see Table 4.3). In Subtask B, **TiGRO-B3** ranked fourth (#4) out of nine submissions under the official soft scoring metric.

4.2.1 Subtask A

Table 4.3 reports the results for Subtask A on the development and test sets. Our best-performing model on the development set. **TiGRO-A1** (micro-F1 57.40), did not generalize well to the test set, exhibiting a significant drop in performance (micro-F1 48.65, rank #5), likely due to overfitting. In contrast, **TiGRO-A3**, our weakest model on the development set (micro-F1 53.81), improved in the test set, achieving the highest micro-F1 of **56.39** among all participant systems.

Table 4.3: Results for Subtask A (*post-level fallacy detection*) on development and test sets. Best result per evaluation set in bold.

	Development				Test				
Model	P	R	micro-F1	macro-F1	P	R	micro-F1	macro-F1	Rank
TiGRO-A1	57.84	57.03	**57.40**	31.58	62.52	39.85	48.65	20.57	5
TiGRO-A2	55.19	53.78	54.46	32.03	53.43	51.48	52.41	27.94	3
TiGRO-A3	49.87	58.47	53.81	32.71	53.24	59.99	**56.39**	33.35	1
MVML-UMB	32.21	11.37	16.8	3.44	38.53	14.28	20.84	5.1	

Table 4.4 shows **TiGRO-A3**'s per-class scores on the test set, averaged across annotators. As argued in Section 4.3, performance highly reflects the class imbalance in the training set, raising concerns about the actual understanding of the system.

Table 4.4: TiGRO-A3 per-class performance on test set, averaged across annotators, ordered by F1 score, with the percentage of training posts.

Fallacy (tag)	Precision	Recall	F1	Train (%)
Appeal-to-emotion (AE)	64.04	91.86	75.47	56.60
Loaded-language (LL)	58.33	92.16	71.26	60.68
Doubt (DO)	65.38	69.19	67.19	18.58
Name-calling-or-labelling (NC)	61.17	68.95	64.73	31.42
Slogan (SL)	72.73	57.14	64.0	11.28
Vagueness (VA)	50.0	73.94	58.9	46.35
Ad-hominem (AH)	56.9	48.53	52.37	12.15
Flag-waving (FW)	41.3	32.24	36.21	13.72
Hasty-generalization (HG)	34.38	37.71	35.91	19.53
Evading-the-burden-of-proof (EP)	29.76	33.55	31.53	17.71
Thought-terminating-cliches (TC)	28.57	26.7	27.59	11.98
Slippery-slope (SS)	37.5	18.75	25.0	7.29
False-analogy (FA)	26.92	13.95	18.18	10.85
Red-herring (RH)	17.39	14.67	15.88	11.20
Cherry-picking (CP)	25.0	7.14	11.11	5.03
Appeal-to-authority (AA)	10.0	4.95	6.6	8.42
Causal-oversimplification (CO)	10.0	3.33	5.0	5.90
Strawman (ST)	0.0	0.0	0.0	5.03
False-dilemma (FD)	0.0	0.0	0.0	3.39
Circular-reasoning (CR)	0.0	0.0	0.0	0.87

4.2.2 Subtask B

Table 4.5 reports Subtask B results on the development and test sets under both strict and soft evaluation. **TiGRO-B3** is the best system across all conditions, achieving a soft micro-F1 of **44.98** and a strict micro-F1 of **42.13** on the test set. Notably, under strict scoring, which provides no partial credit for class mismatches, TiGRO systems occupied the top three positions (**TiGRO-B3**, **TiGRO-B1**, and **TiGRO-B2**), maintaining the same internal performance order observed in the soft scoring (see Table 4.5). The top three ranks in the soft scoring were held by the PuDy team [2], who specifically leveraged the hierarchical fallacy taxonomy (See Section 3.5.3). While we were aware that soft scoring accounts for class relationships, we did not explicitly incorporate this hierarchy information into our development.

Table 4.5: Results for Subtask B (*span-level fallacy detection*) on development and test sets, using micro averaged scores. Best result per evaluation set in bold.

	Development						Test							
	Strict			Soft			Strict				Soft			
Model	P	R	micro-F1	P	R	micro-F1	P	R	micro-F1	Rank	P	R	micro-F1	Rank
TiGRO-B1	37.61	32.37	34.79	41.87	36.80	39.17	38.33	40.35	39.30	2	42.50	45.20	43.80	5
TiGRO-B2	37.59	33.14	35.20	41.50	37.20	39.24	38.23	40.05	39.11	3	42.68	44.87	43.74	6
TiGRO-B3	44.45	31.95	**37.18**	47.81	35.13	**40.50**	47.82	37.67	**42.13**	1	51.01	40.25	**44.98**	4
MVMD-UMB	90.49	0.77	1.53	90.49	0.77	1.53	60.94	3.05	5.80		65.97	3.28	6.25	

Table 4.6 shows **TiGRO-B3**'s per-class scores on the test set, averaged across annotators, together with span frequency and average span length in our unified version of the train-dev set. In this case, span frequency appears to be less predictive of model performance, whereas span length seems to play a more important role, as described in Section 4.3.

Table 4.6: TiGRO-B3 per-class performance on test set, averaged across annotators, ordered by F1 score, with the percentage of training spans and average span length.

Fallacy (tag)	Precision	Recall	F1	Train (%)	Avg span length
Slogan (SL)	71.67	70.93	71.28	2.73	4.32
Doubt (DO)	56.95	75.21	64.80	4.02	17.16
Appeal-to-emotion (AE)	53.94	53.93	53.90	17.85	5.74
Name-calling-or-labelling (NC)	68.51	43.67	53.27	9.84	2.71
Loaded-language (LL)	47.37	46.66	46.90	23.12	2.77
Thought-terminating-cliches (TC)	34.12	37.18	35.55	2.52	5.44
Slippery-slope (SS)	80.00	21.59	34.00	1.54	10.70
Vagueness (VA)	34.57	32.03	32.66	13.03	9.56
Flag-waving (FW)	47.22	21.62	29.66	3.46	4.66
Ad-hominem (AH)	44.87	18.25	25.91	2.62	17.07
Evading-the-burden-of-proof (EP)	28.03	10.25	15.01	3.97	16.52
Appeal-to-authority (AA)	41.67	7.69	12.90	1.90	6.19
Cherry-picking (CP)	48.48	7.14	12.45	1.03	29.21
False-analogy (FA)	26.79	7.19	11.27	2.23	20.93
Hasty-generalization (HG)	18.52	7.85	11.03	4.38	11.01
Red-herring (RH)	11.11	3.49	5.30	2.48	12.79
False-dilemma (FD)	0.00	0.00	0.00	0.80	16.96
Causal-oversimplification (CO)	0.00	0.00	0.00	1.28	20.78
Strawman (ST)	0.00	0.00	0.00	1.03	37.79
Circular-reasoning (CR)	0.00	0.00	0.00	0.18	28.90

4.3 Discussion

In this section, we discuss the results of our models, focusing on our best systems for both subtasks, **TiGRO-A3** and **TiGRO-B3**.

4.3.1 Subtask A

Per-class Analysis

Although we achieved competitive scores overall, a per-class analysis (see Table 4.4) reveals substantial variation in predictive capacity across fallacy types.

Performance broadly reflects the skewed class distribution in the training set: dominant classes such as *Appeal to Emotion* (56.60% of posts) achieve high F1 scores (75.47), while the least frequent classes, such as *Circular Reasoning* (0.87%), are not predicted at all (F1 = 0.0). This is a well-known challenge in multi-label classification under class imbalance: more training instances allow the model to learn more robust patterns, whereas very few examples make generalization nearly impossible. Figure 4.2 illustrates this trend through a scatter plot of per-class F1 score against training frequency, making the overall positive correlation clearly visible.

However, class frequency is not the only factor at play: the intrinsic difficulty of each fallacy type also matters. Dominant classes tend to consist of shorter fallacies rooted in language use [8], such as *Loaded Language*, *Name Calling or Labeling*, and *Appeal to Emotion*, which can often be identified from surface-level lexical cues. In contrast, minority classes mostly correspond to longer, structurally complex logical fallacies [8], such as *Strawman*, *Cherry Picking*, and *Circular Reasoning*, which depend on reasoning over the full argument and cannot be reliably detected from local textual signals. This distinction is illustrated in Examples 1 and 2 from the FAINA dataset: the fallacious nature of the *Loaded Language* instance is signaled by a single identifiable expression (highlighted in bold), while the *Strawman* instance requires understanding the implicit argument distortion spanning the entire post.

(1) Al posto di andare in piazza a **rompere i coglioni** per il Green pass, fatevi fare il vaccino e che sia finita.
(en: *"Instead of going out in the streets to be a* ***pain in the ass*** *about the Green Pass, just get the vaccine and move on."*)

(2) [USER] propone di distribuire i vaccini "anche in base al Pil regionale". In pratica, per lei le regioni più ricche dovrebbero essere più protette dal virus rispetto a quelle più povere. Quale sarà la prossima idea? Vaccinare i bianchi e non i neri? Vergogna.
(en: *"[USER] proposes distributing vaccines 'also based on regional GDP'. In short, for her, richer regions should be better protected from the virus than poorer ones. What will the next idea be? Vaccinate white people but not black people? Shame."*)

The dot size in Figure 4.2 encodes the average span length of each class as a proxy of difficulty. The overall pattern is clear: classes that are both frequent and short-spanning achieve the best results, while those that are rare and long-spanning cluster at the bottom-left of the plot.

Some exceptions are worth discussing. *Doubt* (DO, F1 = 67.19, avg. span 16.1 tokens, 18.58% of posts) and *Ad Hominem* (AH, F1 = 52.37, avg. span 16.0 tokens, 12.15% of posts) have considerably longer average span lengths and modest training frequencies, yet both achieve relatively strong F1 scores, suggesting that despite their length and limited supervision they carry strong signatures that provide reliable surface cues. In the opposite direction, *Appeal to Authority* (AA, F1 = 6.60, avg. span 6.4 tokens, 8.42% of posts) is a short-spanning class of comparable frequency yet performs poorly, suggesting that span length alone is not sufficient when training signal is scarce and the fallacy requires contextual grounding beyond the immediate text. *Slogan* (SL, F1 = 64.0, avg. span 3.5 tokens, 11.28% of posts), by contrast, is similarly infrequent but achieves substantially higher F1: slogans are typically brief and carry strong semantic cues that make them recognizable even from a small number of training examples.

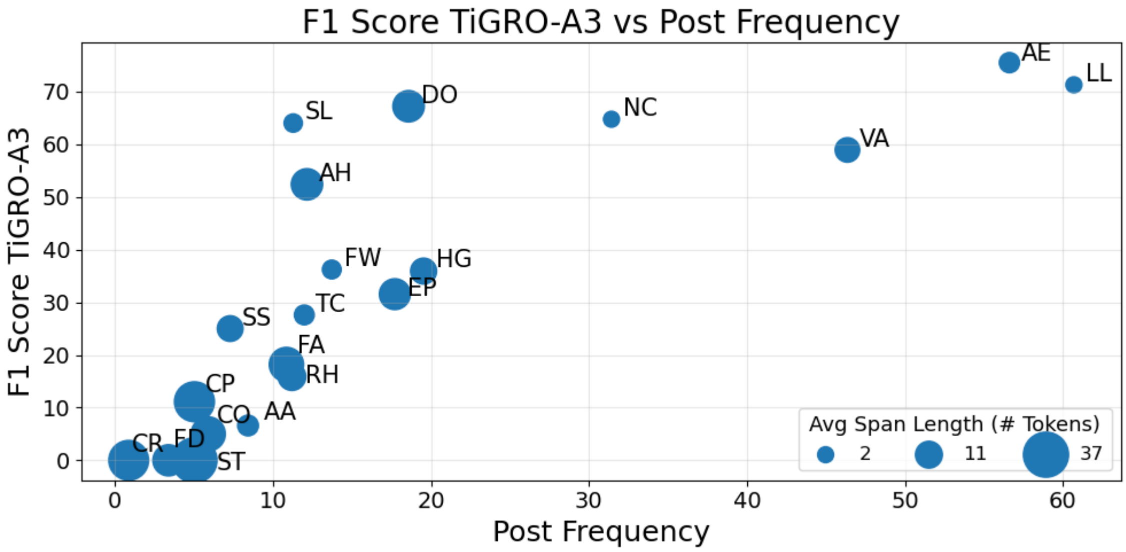


Figure 4.2: Per-class F1 of **TiGRO-A3** on the test set (y-axis) against percentage of training posts per class (x-axis). Dot size encodes average span length.

SVM Comparison These observations raise broader concerns about the reliability of micro-F1 as an evaluation metric in the presence of highly skewed class distributions. Since micro-F1 aggregates performance across all instances, it is inherently dominated by the most frequent classes, while macro-F1 weights each class equally. The competitive micro-F1 scores achieved by our systems in the shared task ranking may overstate its actual capabilities, masking poor performance on the majority of fallacy classes.

To further assess whether the strong performance on frequent classes might be due mostly to exploitable surface patterns rather than genuine semantic understanding, we trained a linear *Support Vector Machine* (SVM) as a simple baseline, using the `scikit-learn` implementation with default hyperparameters [103] and sentence embeddings from `xlm-r-distilroberta-base-paraphrase-v1` [104]. Since the test set gold labels were not released, we report results on the development set to ensure a fair comparison.

The SVM achieves a micro-F1 of 47.38, remarkably close to **TiGRO-A3**'s 53.81 on the same set (Table 4.3). However, macro F1 provides a fairer assessment of model capabilities with **TiGRO-A3** achieving 32.71 compared to only 15.04 for the SVM, suggesting that the encoder-based model captures a broader range of class-specific signals beyond simple pattern matching.

More revealing insights emerge from the per-class analysis. On the development set, **TiGRO-A3** achieves a per-class F1 of at least 50% for only 5 out of 20 fallacy classes, and crucially, these are precisely the same five classes for which the SVM obtains a non-zero F1 score. Table 4.7 reports development set scores for both models on these top five classes. These classes are among the most frequent in the training set and tend to rely on identifiable surface-level lexical patterns: *Doubt*, *Loaded Language*, *Appeal to Emotion*, *Name Calling*, and *Vagueness*. The similarity of their scores strongly suggests that these classes are inherently easier to detect even without complex context understanding, possibly due to exploitable lexical biases in the dataset. In contrast, the SVM fails entirely on all remaining classes, while **TiGRO-A3** manages to detect several mid-difficulty classes, albeit with limited performance. This partial advantage confirms that the encoder-based model captures some patterns beyond surface cues, yet it remains difficult to reliably assess how well it actually understands complex fallacies.

Table 4.7: Performance of SVM and **TiGRO-A3** on the five fallacy classes for which SVM achieves non-zero F1 on the development set. All remaining classes achieve F1 = 0 for the SVM and below 50% for TiGRO-A3.

	SVM			TiGRO-A3		
Fallacy (tag)	P	R	F1	P	R	F1
Doubt (DO)	89.58	54.56	67.79	73.75	74.84	74.26
Loaded-language (LL)	54.24	82.59	65.30	55.90	91.71	69.28
Appeal-to-emotion (AE)	58.03	72.59	64.47	57.52	80.43	67.04
Name-calling-or-labeling (NC)	52.56	39.87	45.33	51.35	73.85	60.57
Vagueness (VA)	50.46	68.95	57.89	48.39	75.37	58.56

Confusion Matrix Table 4.8 reports per-class confusion matrix counts for **TiGRO-A3** on the development set. With 20 classes in a multi-label setting, a single confusion matrix is not well-defined, so we limit our analysis to the aggregated TP, FP, FN, and TN counts per class. As expected, dominant classes accumulate the highest true positive counts: *Loaded Language* (121 TP), *Appeal to Emotion* (101 TP), and *Vagueness* (74 TP). For minority classes, absolute counts are so low that performance metrics become unreliable: evaluating *Circular Reasoning* on 3 instances or *False Dilemma* on 7 raises the question of whether meaningful conclusions can be drawn at all.

Beyond this data scarcity issue, the model largely does not attempt to predict rare classes, as reflected in the near-zero TP and high FN counts at the bottom of the table. Notably, *False Analogy* (3 TP, 14 FP) and *Red Herring* (2 TP, 15 FP) show disproportionately high false positives, suggesting the model fires on these labels but systematically confuses them with other classes.

Table 4.8: TiGRO-A3 confusion matrix counts per fallacy type on the development set, ordered by true positive count.

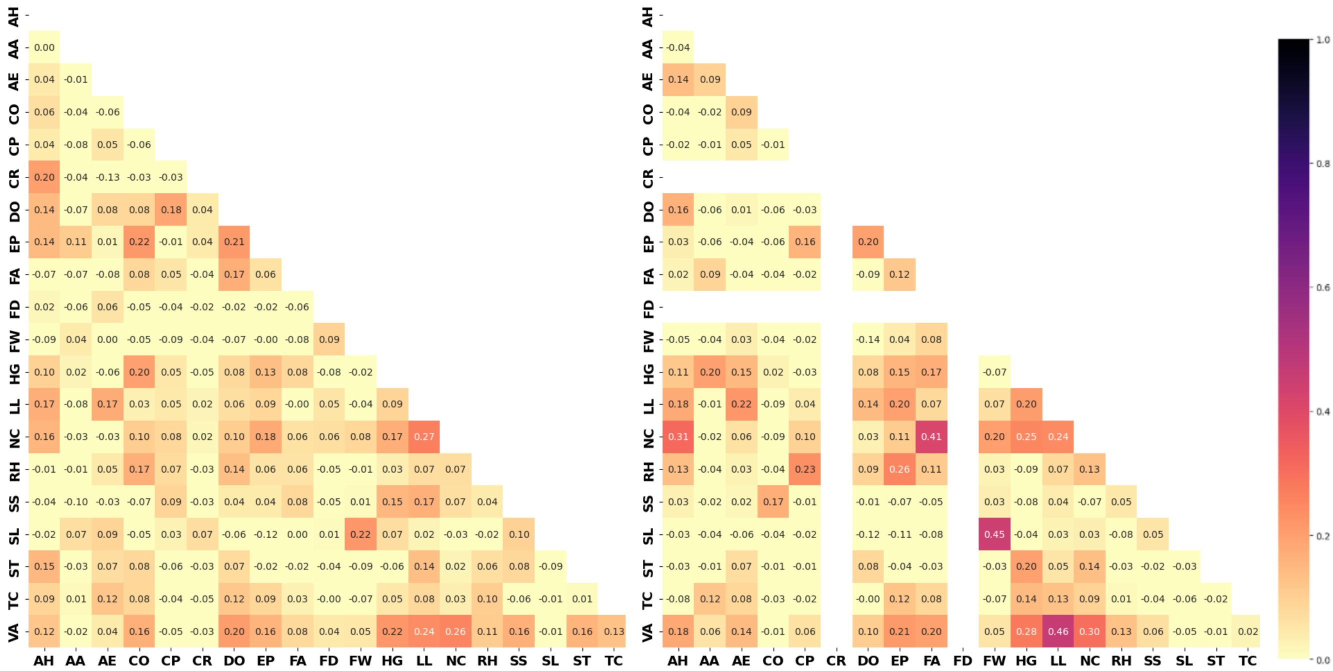
Fallacy (tag)	TP	FP	FN	TN
Loaded-language (LL)	121	57	11	42
Appeal-to-emotion (AE)	101	52	27	51
Vagueness (VA)	74	50	30	77
Name-calling-or-labelling (NC)	44	30	18	139
Doubt (DO)	31	9	13	178
Evading-the-burden-of-proof (EP)	15	19	29	168
Hasty-generalization (HG)	15	27	24	165
Flag-waving (FW)	12	9	15	195
Slogan (SL)	11	5	19	196
Ad-hominem (AH)	10	12	16	193
Thought-terminating-cliches (TC)	7	5	26	193
Slippery-slope (SS)	4	3	12	212
Appeal-to-authority (AA)	3	1	23	204
False-analogy (FA)	3	14	20	194
Red-herring (RH)	2	15	17	197
Causal-oversimplification (CO)	1	3	15	212
Strawman (ST)	0	2	13	216
Cherry-picking (CP)	0	1	12	218
False-dilemma (FD)	0	0	7	224
Circular-reasoning (CR)	0	0	3	228

Fallacy Co-occurrence Matrix The co-occurrence matrices in Figure 4.3 complement this analysis by comparing label associations in the gold annotations and in **TiGRO-A3** predictions. With small counts per class, co-occurrence values can be noisy and prone to overinterpretation, so we focus on the most salient patterns. The predicted matrix broadly reproduces the dominant associations of the gold, particularly the strong co-occurrences among *Loaded Language* (LL), *Appeal to Emotion* (AE), *Name Calling* (NC), and especially, *Vagueness* (VA). These classes tend to appear together because they reflect pervasive stylistic features of online discourse rather than structurally isolated fallacy types. However, correlations are generally inflated in the predicted matrix, reflecting a systematic bias toward majority classes.

Two specific confusions stand out. First, *Flag Waving* (FW) and *Slogan* (SL) are over-predicted as co-occurring, which is unsurprising given that both exploit group identity and rhetorical force, as in Example 3, and are described as closely related types (see Section 1.5). Second, *False Analogy* (FA) shows an inflated correlation with *Name Calling* (NC), possibly because drawing a false analogy often involves implicitly *"labeling"* a target through comparison, as in Example 4.

(3) [...] Difendiamo i confini #BloccoNavaleSubito
(en: *"[...] Let's defend the borders #NavalBlockadeNow"*)

(4) La ministra apre i porti ai **migranti** affetti da COVID e se ne frega dei due **senzatetto** morti a Milano.
(en: *"The minister opens the ports to* ***migrants*** *with COVID and doesn't give a damn about the two* ***homeless*** *people who died in Milan."*)



(a) Gold labels, development set.

(b) Predicted labels (**TiGRO-A3**).

Figure 4.3: Correlation of fallacy label co-occurrences in the development set (a) and in **TiGRO-A3** predictions (b), considering merged annotations.

Performance per Annotator During our experiments on the development set, we observed that results for the second annotator are consistently higher across all setups, including Subtask B. This is also confirmed on the test set: **TiGRO-A3** achieves a micro-F1 of 58.68% on Annotator 2 compared to 54.10% on Annotator 1, and similarly, **TiGRO-B3** scores 46.33% on Annotator 2 against 43.63% on Annotator 1 under soft evaluation. This asymmetry may be attributable to Annotator 1 providing a noisier signal, possibly due to a less consistent labeling style, making Annotator 2's annotations inherently easier to model. Merging the two annotation sets into their union, rather than training on each independently, likely helped stabilize the training signal and improve overall performance.

4.3.2 Subtask B

Per-class Analysis As shown in Table 4.6, performance still broadly reflects class distribution, but span frequency turns out to be a weaker predictor than in Subtask A, while average span length emerges as a more dominant factor. Figure 4.4 illustrates this with a scatter plot of per-class F1 score against average span length, with dot size representing span frequency, revealing a clear negative correlation: shorter-span classes cluster in the upper-left region of the plot, while longer-span classes are consistently pushed toward zero regardless of their frequency. Intuitively, a fallacy that spans an entire post offers no reliable surface anchor for boundary detection, as in *Strawman* (ST, avg. 37.79 tokens, F1 = 0.0) and *Circular Reasoning* (CR, avg. 28.90 tokens, F1 = 0.0). Conversely, short-span classes benefit from the relative ease of localizing a compact, lexically distinctive trigger; *Slogan* (SL) is the clearest example, achieving F1 = 71.28 while representing only 2.73% of training spans and having an average span length of just 4.32 tokens.

Two exceptions are worth discussing. *Appeal to Authority* (AA) performs poorly (F1 = 12.90) despite a relatively short average span length of 6.19 tokens. However, given its even weaker performance in Subtask A (F1 = 6.60), the class appears to be inherently difficult, possibly due to the low training frequency.

Conversely, *Doubt* (DO) achieves F1 = 64.80 despite an average span length of 17.16 tokens, the longest among the well-performing classes. Its strong performance is consistent across both subtasks and is matched even by the SVM (Table 4.7), suggesting that *Doubt* is inherently the easiest class to detect despite its relatively low training frequency (4.02% of spans, 18.58% of posts). This may be explained by strong semantic cues, most notably its characteristic interrogative syntactic structure, as in Example 5.

(5) Ho fatto il vaccino ma non mi sono fatto fotografare, funziona comunque?
(en: *"I got the vaccine but didn't take a photo doing it, does it still work?"*)

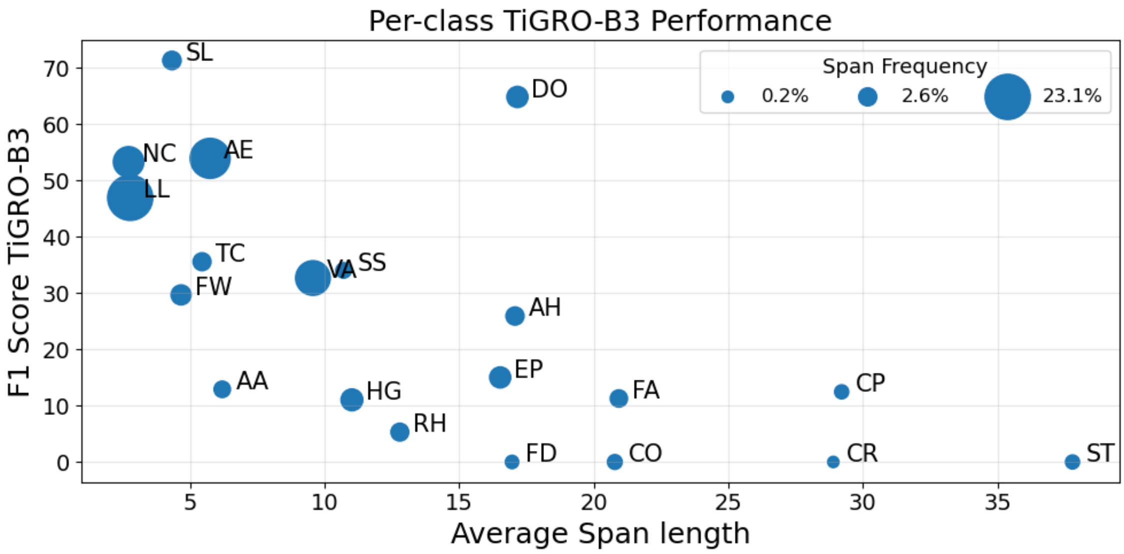


Figure 4.4: Per-class F1 score of **TiGRO-B3** on the test set (y-axis) against average span length (x-axis). Dot size encodes span frequency in the training set.

Subtasks comparison Overall, Subtask B is considerably more challenging than Subtask A, as it requires not only recognizing the fallacy type but also precisely localizing it in the text, resulting in lower performance across almost every class. The class ordering also shifts between the two subtasks. *Hasty Generalization* (HG, 35.91 → 11.03) and *Evading the Burden of Proof* (EP, 31.53 → 15.01) drop sharply, indicating that while their post-level presence is relatively detectable, identifying the exact span is harder when the fallacy is diffuse across the post rather than anchored to a few specific words. One notable exception is *Slogan* (SL, 64.0 → 71.28), which improves substantially in Subtask B, consistent with its short, localized nature.

These results confirm that span-level fallacy detection remains an open challenge, particularly for complex logical fallacies that lack identifiable surface anchors and are hard to detect even at the post level. As discussed in Chapter 5, improving performance on such classes will likely require both higher-quality training data and richer modeling of discourse-level structure.

4.4 Methods not Adopted

During system development, we experimented with several alternative approaches to improve performance. While not all succeeded, discussing these attempts may benefit future work.

4.4.1 Data Integration

To address class imbalance (Section 3.3.3), a common approach is to extend the training set with external resources or synthetically generated examples. We experimented with this direction early in system development, but ultimately abandoned it due to inconclusive results.

Specifically, we attempted to expand the training data by integrating examples from external datasets. This is a well-known challenge in fallacy detection: as argued in Chapter 1, the lack of shared fallacy definitions has produced a vast literature of fundamentally different datasets and annotation guidelines, making integration inherently difficult. We decided to use the **MAFALDA** dataset introduced by Helwe et al. [36], since its fallacy classes are partially aligned with FAINA's, making label mapping feasible. MAFALDA consists of 200 annotated English examples drawn from diverse sources, including Reddit discussions, toy examples, and political debates.

After harmonizing the labels, we first incorporated the English MAFALDA examples directly into the training set for Subtask A, following the same multi-task architecture as TiGRO-A2 but using mmBERT as the backbone encoder to leverage its multilingual capabilities. This did not yield noticeable improvements. As a second attempt, we automatically translated the MAFALDA examples into Italian using Claude Sonnet 4.5, allowing us to use the better-performing ALBERTo encoder. This approach, however, led to a drop in performance. We attribute this degradation to two main factors: (i) *domain mismatch*, as MAFALDA examples differ substantially in register and style from Twitter posts; and (ii) *annotation guideline divergence*, despite the harmonization effort, differences in fallacy class definitions introduced noise into the training signal. Given these discouraging results, we did not pursue further data integration strategies.

Our experience is consistent with the broader findings from other FadeIT participants, who attempted data augmentation through synthetic example generation, also obtaining mixed results [2]. MALTO used ChatGPT to generate paraphrases of underrepresented training instances; PuDy used Gemini to perturb the context surrounding annotated spans; and UNICA generated additional minority-class examples using GPT-5.1, and via back-translation through Spanish and French as pivot languages. While PuDy's context perturbation approach appeared promising, MALTO's paraphrasing strategy led to performance degradation, likely due to label noise introduced during generation, and UNICA's combined strategy yielded inconsistent gains.

Taken together, these findings suggest that data augmentation alone is unlikely to resolve class imbalance in the absence of higher-quality, in-domain training data. We believe the most impactful long-term solution lies in collecting and annotating more diverse, in-domain data, particularly for the rarest fallacy types, as argued in Chapter 5.

4.4.2 Span Extraction with Attribution

For addressing Subtask B, our initial idea was to build a performant post-level classifier for the 20 fallacy types (Subtask A), followed by span extraction using attribution methods (see Section 2.4).

Motivation The idea behind this approach is straightforward: if a classifier correctly predicts that a post contains a given fallacy type, attribution methods can be used to identify which tokens most influenced that decision. These token-level relevance scores can then serve as a proxy for span localization, under the strong assumption that the tokens the model relies on to recognize a fallacy correspond to its fallacious portion in the text.

This approach is appealing because it does not require any span-level supervision during training and directly reuses the post-level classification head for each fallacy. The design is illustrated in Figure 4.5.

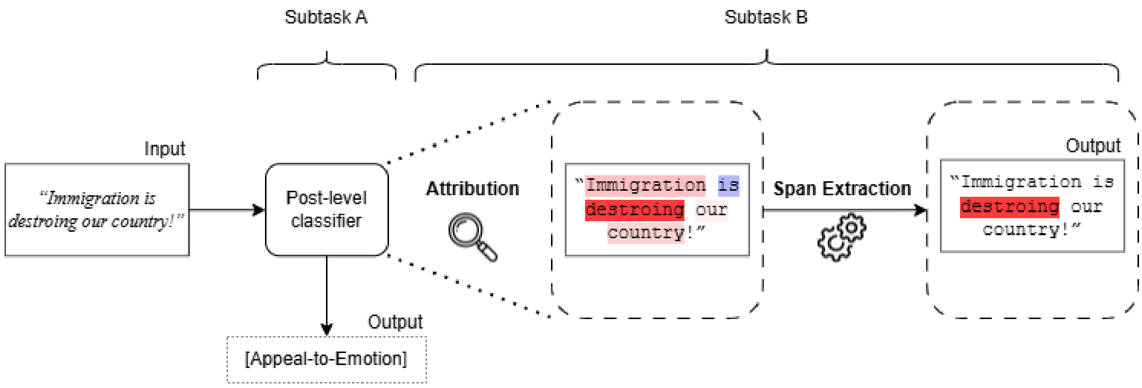


Figure 4.5: Overview of the attribution-based span extraction pipeline. A post-level classifier predicts the fallacy type; attribution scores are then used to identify the most relevant tokens, from which a span is extracted.

Feasibility In practice, this approach depends critically on the quality of the upstream post-level classifier. Due to weak performance on Subtask A across most classes (Section 4.2), it was not generally feasible: a classifier that does not reliably detect a fallacy at the post level will produce attribution maps that are at best uninformative and at worst actively misleading. For this reason, the approach was abandoned as a general solution for Subtask B, but we still experimented with the highest-performing class, *Appeal to Emotion*, as an exploratory proof of concept.

Method We used the `lxt` library, which provides an efficient implementation of AttnLRP [74], to compute a relevance score for each input token, indicating how positively or negatively it contributes to the predicted class. Since the encoder operates on subword tokens, we reconcile its tokenization with the annotation-level tokenization by assigning each word the maximum relevance among its constituent subwords, treating the whole word as the unit of attribution. Spans are then extracted by selecting all tokens whose relevance exceeds a threshold, defined as θ standard deviations above the mean relevance across the post. The value of θ was tuned on the development set to maximize macro-F1 under BIO token-level evaluation for the considered class.

Results and Analysis Our preliminary experiments on *Appeal to Emotion* yielded inconclusive results: while attribution proved useful as a human inspection tool, automatic span extraction remained poor overall.

Figure 4.6 illustrates some qualitative examples. Red indicates a positive contribution to the predicted class and blue a negative one; the gold span is highlighted in **bold**. Relevance scores are not always clean and may contain noise with no clear human interpretation. Nevertheless, we can observe that the model generally focuses on emotion-related words that support correct predictions for *Appeal to Emotion*.

- Example 4.6a shows a perfect match: the attribution span aligns exactly with the gold annotation (*disumano*, en: "*inhumane*"), showing that the model can in principle focus on precisely the right portion of the text, although this is not usually the case.

- Example 4.6b illustrates the more typical situation, revealing two main failure. First, the model tends to assign high relevance to single emotionally salient words (*lutto*, en: "mourning") rather than the full annotated span (*in segno di lutto*, en: "as a sign of mourning"). This sparsity in the relevance distribution makes it hard to recall coherent spans of the length expected by the annotation. Second, the model is sometimes distracted by tokens that contribute to the post-level prediction but fall outside the gold span, introducing noise that complicates exact span boundaries. Interestingly, some of those tokens (*vittime innocenti animali*, en: "innocent animal victims") are arguably emotionally charged and could themselves constitute a valid fallacy span, pointing to genuine annotation ambiguity rather than a clear model error.

- Example 4.6c is a false positive. The classifier incorrectly predicts *Appeal to Emotion*, but attribution serves as a useful explainability tool: the highlighted region (*accoglienza per immigrati*, en: "for welcoming immigrants") is interpretable as emotionally relevant, even though in this case it is actually just descriptive.

- Example 4.6d is a false negative. Here the colors are inverted, since the post-level prediction was negative: blue now indicates a positive contribution toward the fallacy presence. The model assigns strong relevance precisely to the gold span (*Vi abbraccio*, en: "sending hugs"), yet the final prediction was still wrong. Attribution thus reveals a case where the internal representations were pointing in the right direction, but something in the final decision failed to reflect that.

Considerations Although our preliminary experiments were inconclusive, we believe attribution-based span extraction remains an interesting direction. Making it more competitive would require designing more sophisticated span extraction heuristics to better handle noisy and sparse relevance distributions, as well as evaluating the approach across other fallacy types with varying span lengths. We leave a more systematic investigation to future work.

Decreto sicurezza : multe fino a 50mila euro e sequestro delle navi per le ONG che salvano migranti in mare . Questo è un governo **disumano** , lordo e criminale . # decretosicurezza # governo # Meloni # criminale
(en: *"Security decree: fines up to 50k euros and seizure of ships for NGOs that rescue migrants at sea. This is an* ***inhumane****, filthy, and criminal government. #securitydecree #government #meloni #criminal"*)

(a) True positive: attribution span matches the gold span exactly.

Buongiorno a tutti , per rispetto ai tanti sfollati , alla **terribile** emergenza in Sardegna , al **disastro** ecologico , alle vittime innocenti animali , oggi **in segno di lutto** posterò solo informazioni riguardanti appelli urgenti e informazioni utili . Grazie Stefania
(en: *"Good morning everyone, out of respect for the many displaced people, the* ***terrible*** *emergency in Sardinia, the ecological* ***disaster****, and the innocent animal victims, today,* ***as a sign of mourning****, I will only post information about urgent appeals and useful updates. Thank you, Stefania"*)

(b) True positive: attribution highlights key emotional words but fails to cover the full gold span; some highlighted tokens fall outside the gold annotation.

In 10 anni tagliati 200 ospedali , 45 mila letti , 10 mila medici e 11 mila infermieri . In 10 anni aperti 9 . 282 centri di accoglienza per immigrati . Nient ' altro da aggiungere .
(en: *"In 10 years, 200 hospitals, 45,000 beds, 10,000 doctors, and 11,000 nurses have been cut. In 10 years, 9 have been opened. 282 centers for welcoming immigrants. Nothing else to add"*)

(c) False positive: attribution reveals the emotionally charged passage that likely triggered the wrong prediction.

Al pronto soccorso . trascorrero ' la nottata in obi , osservazione intensiva . domani ulteriori accertamenti . **vi abbraccio**
(en: *"At the Emergency room. I'll spend the night in OBI, intensive observation. More tests tomorrow.* ***Sending hugs****."*)

(d) False negative: colors are inverted (blue = positive contribution to the gold class); the model focused on the correct span but still predicted incorrectly at the post level.

Figure 4.6: Attribution examples for **TiGRO-A1** predictions on *Appeal to Emotion*. Red indicates a positive contribution to the predicted class, Blue a negative contribution, and the gold span is in **bold**.

Chapter 5

Conclusions

This thesis presented **TiGRO**, a multi-task learning approach to automatic fallacy detection developed for the **FadeIT** shared task at EVALITA 2026. Our systems achieved competitive results across both subtasks: **TiGRO-A3** ranked first in Subtask A (*post-level fallacy detection*) with a micro-F1 of **56.39**, while **TiGRO-B3** ranked fourth in Subtask B (*span-level fallacy identification*) under soft evaluation (micro-F1: **44.98**) and first under strict evaluation (micro-F1: **42.13**).

The core of our approach, described in Chapter 4, is the joint modeling of both subtasks through multi-task learning. Rather than treating post-level classification and span-level identification as independent problems, our best-performing systems fine-tuned a single shared encoder over 40 parallel tasks: 20 binary classification heads for Subtask A and 20 BIO sequence labeling heads for Subtask B. This design allowed the model to leverage complementary signals across tasks and proved consistently effective across encoder choices and evaluation conditions.

While these results demonstrate the value of multi-task learning within the shared task setting, they do not necessarily translate into a genuine understanding of fallacious reasoning. The remainder of this chapter attempts to provide a broader perspective on the limitations of our work in the general context of automatic fallacy detection (AFD).

5.1 Critical Review

Although our systems achieved competitive micro-F1 scores, a deeper analysis of our results, detailed in Section 4.3, gives a different picture. **TiGRO-A3** ranked first on micro-F1 (56.39), but per-class analysis (Table 4.4) shows that performance is highly uneven, closely following the skewed class distribution of the dataset. The model achieves strong performance only on the dominant classes (e.g., *Appeal to Emotion*, *Loaded Language*, *Doubt*, *Name Calling*), while performing poorly on all others and failing completely on the minority classes (e.g., *Strawman*, *False Dilemma*, *Circular Reasoning*). Unfortunately, the extreme class imbalance makes it hard to draw meaningful conclusions about minority class performance: the risk is that these classes are not so much *undetected* as they are *undetectable* from the available training data alone.

To assess how much of this performance reflects genuine understanding versus surface pattern matching, our experiment with a simple SVM baseline (Section 4.3) offers valuable insight. On the development set, the SVM achieves close performance on exactly the same top five classes as **TiGRO-A3** (Table 4.7), while failing to detect all others. This suggests that the dominant classes are inherently easier to detect, relying on superficial cues and possibly on dataset bias, which makes them less interesting from the broader perspective of fallacy detection. However, these considerations would have been masked by micro-F1 alone: the SVM achieves a surprisingly strong micro-F1 of 47.38, close to **TiGRO-A3**'s 53.81 on the same set.

This is precisely why we advocate for macro-F1 as the primary evaluation metric for fallacy detection: it gives equal weight to all classes, which is especially important given that the rarest classes are also the most difficult and arguably the most valuable to detect. Under this perspective, the gap between the two models becomes much more meaningful: the SVM's macro-F1 of 15.04 against **TiGRO-A3**'s 32.71 shows that the encoder-based model does capture something beyond surface cues, even if performance remains unsatisfying. More broadly, using micro-F1 for ranking in the shared task overstates the actual capabilities of all submitted systems: we have optimized for what is measurable, the frequent classes, while the harder, rarer fallacy types remain largely undetected.

In fact, the shared task overview [2] shows that the **UNICA** [100] system, based on a retrieval-augmented generation pipeline with LLMs and dynamic few-shot prompting (Section 3.5.3), achieved the strongest macro-F1 (36.75) on the test set, exceeding **TiGRO-A3** (33.35), with more balanced scores across classes largely due to better performance on minority fallacy types. This aligns with recent evidence that LLMs can generalize more robustly to rare classes [38, 41], albeit at the cost of weaker performance on the dominant classes where surface patterns are more exploitable by supervised learning. In contrast, supervised AFD models tend to rely heavily on the distributional properties of their training set and struggle to generalize beyond them. Our results provide further empirical confirmation of this claim.

5.2 Dataset limits

Many of the limitations observed in our models reflect deeper constraints of the FAINA dataset itself, rather than failures of the modeling approach. FAINA represents important progress in bringing perspectivist annotation to Italian social media, but its structural properties constrain what supervised models can learn from it.

Class imbalance As previously discussed, the most fundamental limitation of FAINA is its **highly skewed** class distribution (Table 3.2), which poses a severe challenge for supervised learning. With *Loaded Language* appearing in 60.7% of posts and *Circular Reasoning* in only 0.87%, the most complex classes are also the most underrepresented. The scarcity of training signal for these classes makes generalization nearly impossible, as clearly reflected in the per-class performance results (Tables 4.4 and 4.6), where several minority classes receive near-zero or zero F1 scores across both subtasks.

This imbalance is not accidental: it is a direct consequence of using Twitter/X as the data source, which introduces a systematic bias toward the fallacy types that naturally occur in short-form, emotionally charged social media discourse. Character limits and viral dynamics favor brief, rhetorically punchy expressions over extended logical reasoning, making language-based fallacies such as *Loaded Language* or *Appeal to Emotion* far more common than structural ones like *Strawman* or *False Dilemma*. We understand the appeal of Twitter/X as a data source, particularly for Italian, where large annotated corpora are scarce. However, we question whether this platform is the right domain for a task where the most challenging and interesting fallacy types are structurally suppressed by the medium itself. More structured domains could provide a richer environment for complex argumentation while remaining accessible for the Italian language, and socially relevant, such as Reddit discussions [47], political debates [33], or news articles [44], all of which are already represented in the broader AFD literature.

Insufficient annotators A central design principle of FAINA is its perspectivist approach (Section 3.2), which retains individual annotation perspectives rather than collapsing them into a single ground truth. However, with only **two annotators**, this philosophy is underrealized. As discussed in Section 3.2, fallacy identification depends on multiple factors that vary across individuals: interpretation of speaker intent, familiarity with the topic, cultural and linguistic background, and personal thresholds for what constitutes a fallacy. Two perspectives are insufficient to capture the genuine diversity of interpretations. Cases of genuine ambiguity are resolved as binary disagreements rather than as distributions over plausible readings, and a single noisy annotator can skew the entire training signal.

Annotation quality Beyond the structural issue of having too few annotators, FAINA also exhibits specific instances of annotation **inconsistency** and ambiguity that complicate model training and evaluation. This is partly because fallacy definitions are difficult to apply to everyday discourse, particularly on social media, where posts rarely contain fully formed arguments. The result is a dataset where some instances do not always serve as clear representatives of their assigned fallacy class. We emphasize that this is not a critique of the annotators' effort; however, acknowledging these limitations is nonetheless important for interpreting model performance.

To further illustrate our point on annotation quality and the need for multiple annotator perspectives, we present a qualitative overview of instances from the dataset whose annotation is arguably problematic, precisely the kind of cases discussed in Section 3.2.

- **Example 6** is a well-known Italian saying that was appropriately annotated as *Appeal to Emotion* (AE) and *Thought-Terminating Cliché* (TC), as it contributes no substantive argument. However, it was additionally labeled as *Circular Reasoning* (CR) and *False Dilemma* (FD). While one can suppose a plausible interpretation motivating these labels, the post does not present any argument at all, making these assignments questionable. This is even more notable, considering that this is one of the few instances of *Circular Reasoning*.

(6) Ci sarebbe da ridere se non ci fosse da piangere. [AE | TC | CR | FD]
(en: *"It would be funny if it weren't so sad."*)

- **Example 7** was appropriately annotated with *Vagueness* (VA) and *Evading the Burden of Proof* (EP), but also received a *Loaded Language* (LL) label triggered by the word *governare* (en: *managing*). Since loaded language is defined as language carrying strong connotations, this assignment suggests a particularly low annotation threshold.

(7) Rimettete Salvini al Viminale, è l'unico capace di **governare** l'arrivo dei migranti. [VA | EP | LL]
(en: *"Put Salvini back at the Interior Ministry, he's the only one capable of* ***managing*** *the arrival of migrants."*)

- **Examples 8 and 9** form a contrastive pair illustrating how domain knowledge affects annotation consistency. In the first example, the hashtag *#UnaVocePerSanMarino* (simply the name of a well-known song contest) was labeled as a *Slogan* (SL). In contrast, the second example contains *#PianoRimpatriSicuri* (en: *#SafeRepatriationPlan*), a known political slogan, which received no label from either annotator. This discrepancy is possibly attributed to differing levels of familiarity with the relevant cultural and political contexts.

(8) La sala di emissione di RTV #UnaVocePerSanMarino [SL]
(en: *"The RTV broadcasting room #UnaVocePerSanMarino"*)

(9) Questa mattina dal [USER] presentiamo il #PianoRimpatriSicuri [-]
(en: *"This morning from [USER] we present the #SafeRepatriationPlan"*)

- Finally, **Example 10** is a particularly subtle case. The post was annotated as non-fallacious; however, familiarity with niche social media knowledge reveals that the raising-hand emoji (🙋‍♀️) is used in certain online communities to evoke the fascist salute, lending a charged meaning to a post about vaccine side effects during the COVID-19 pandemic. Under this reading, the post could plausibly be labeled as *Loaded Language*, *Appeal to Emotion*, *Vagueness*, or *Thought-Terminating Cliché*.

(10) Postumi del vaccino... Braccio 🙋‍♀️ [-]
(en: *"Vaccine side effects... Arm* 🙋‍♀️*"*)

5.3 Future Directions

The limitations identified above reduce to three fundamental challenges in AFD: (i) the number of fallacy types is large and datasets are highly imbalanced; (ii) existing datasets cover heterogeneous genres and are typically small due to the high cost of annotation; and (iii) models trained on individual datasets often exhibit poor out-of-distribution generalization [38, 37]. The remainder of this section discusses concrete directions for future work to address them.

Improving Datasets The most impactful long-term investment for advancing AFD is the creation of larger, more diverse, and more carefully annotated datasets. While technical innovations in model architecture and training procedures can yield incremental gains, they cannot overcome the fundamental bottleneck of insufficient and poor-quality training data. To improve annotation quality, we find particularly compelling the annotation scheme introduced by Helwe et al. [36] in the MAFALDA dataset. Rather than relying solely on informal definitions, they also provide formal fallacy definitions inspired by Bennett [105] and also deployed in other works [38]. Despite the difficulty of providing a real formal definition of fallacies within a strict logical framework, as discussed in Chapter 1, they adopt a pragmatic approach, offering structured templates whose primary goal is to provide effective annotation guidelines, rather than real theoretical grounding. For each fallacy, the template specifies the relationships between *entities*, *premises*, and *conclusions*. Annotators are required not only to label a span, but also to justify their annotation by instantiating these template variables. This approach enforces more deliberate reasoning during annotation, reducing errors due to carelessness or misunderstanding while preserving genuine disagreement. This design is particularly well suited to the perspectivist approach to annotation discussed in Chapter 3. Template-based justifications make disagreements explicit and interpretable, allowing multiple valid annotations to be associated with a single instance while minimizing noise. Each annotation perspective, together with its justification, is treated as equally valuable, reflecting genuine interpretive variation. The template structure guides the annotation process and reduces noise, making it more accessible to larger annotator pools and potentially enabling crowd-sourcing, possibly with expert revision. The resulting dataset would provide richer and more reliable labels, combining perspectivist multi-label annotations at the span level with structured justifications for each annotation decision.

Finally, the availability of structured explanations alongside labels would provide an additional training signal for recent promising approaches that aim to enhance forms of logical reasoning in LLMs through fallacy understanding [39, 41]

Multi-Domain Training One of the fundamental limitations of current AFD systems, including ours, is their reliance on dataset-specific biases. As a result, they often fail to generalize when evaluated on data from different domains, genres, or annotation schemes Pan et al. [38]. When models depend too heavily on such biases, simply incorporating additional data from different sources may even degrade performance, as observed in our attempt to merge FAINA and MAFALDA (Section 4.4). However, if the goal is to develop truly general AFD systems, it is necessary to explicitly account for the diversity of domains in which fallacies occur and to design models that incorporate this broader perspective.

In addition to improving datasets, through the construction of more comprehensive resources, as discussed above, a complementary direction is to develop models directly designed to learn from heterogeneous data sources simultaneously.

Our results offer a limited but meaningful indication that multi-task learning can improve performance across subtasks. This suggests that sharing information across fallacy types and levels of granularity may help models capture more general properties of fallacious reasoning. In this sense, our findings are compatible with approaches that extend multi-task learning across datasets.

This perspective is clearly articulated by Alhindi et al. [37], who propose a multi-task, instruction-based prompting framework for fallacy detection. In their approach, each dataset is treated as a separate task, with its own label space, input format, and domain characteristics, and a single T5 model is trained to handle all tasks jointly. Such approaches represent a promising direction for learning more generalizable representations across heterogeneous taxonomies and domains.

In this context, cross-dataset evaluation must become standard practice. Rather than optimizing for a single benchmark, systems should be required to demonstrate robustness across multiple datasets to provide a reliable assessment of their general capabilities, as recently emphasized across the literature [106, 37, 39, 107, 38, 41]. Indeed, the field is converging on a set of core benchmarks that facilitate direct comparison between systems, most notably: ARGOTARIO [108], LOGIC [35], COVID-19 [32], CLIMATE [37], REDDIT [47].

The role of LLMs The role of large language models (LLMs) in Automatic Fallacy Detection (AFD) remains an open and evolving question. Given the intrinsic complexity of AFD, it also represents a particularly interesting task for evaluating the limits of LLM capabilities, especially as they keep saturating benchmarks.

Current evidence points to mixed results: in our experiments, as well as in prior work [38], encoder-based models consistently outperform LLMs on specific in-domain datasets. However, as argued throughout this thesis, strong benchmark performance does not necessarily reflect genuine understanding.

Despite their lower performance in standard supervised settings, LLMs appear to offer promising complementary properties. In particular, they tend to exhibit better generalization across domains and show relatively stronger behavior on minority classes, suggesting a broader and more flexible representation of fallacious reasoning. At the same time, their generative nature introduces new challenges: outputs are harder to control, predictions are less stable, and performance degrades when dealing with large and fine-grained label spaces [101, 37, 2].

Nevertheless, recent work has shown that LLMs can achieve competitive results even in zero-shot settings when guided by carefully designed prompts [38]. This suggests that their apparent limitations is, at least in part, due to suboptimal prompting strategies rather than inherent model weaknesses, and that their full potential is still underexplored. In this direction, the multi-task instruction-based prompting framework discussed earlier [37] provides further evidence that a more structured approach can significantly improve performance, enabling LLMs to better handle heterogeneous definitions and domains.

A promising direction for future work is to leverage the complementary strengths of both paradigms rather than treating them as competing alternatives. One straightforward strategy would be to select the most suitable model for specific subsets of fallacies (e.g., encoder-based models for language-related fallacies and LLMs for more complex logical ones). Conversely, a more comprehensive approach is to design hybrid architectures. Recent work has explored this direction with encouraging results, using LLMs to generate intermediate reasoning steps and extract features (e.g., identifying premises and conclusions or producing explanations), which are then used as structured inputs for downstream encoder models [41, 109].

Investigating the interaction between these approaches and understanding how to effectively combine them, represents a valuable direction for advancing AFD systems beyond current limitations.

5.4 Ethical Statement

This work is intended to contribute to the development of human-in-the-loop assistants that support critical thinking and the identification of fallacious reasoning and misinformation in public discourse. At the same time, we acknowledge that AFD carries important ethical risks.

Classification errors, particularly false positives, may lead to the incorrect flagging of legitimate arguments as fallacious, with potential implications for freedom of expression. In addition, the strong lexical biases present in our models risk reinforcing cultural or linguistic stereotypes embedded in the training data.

We emphasize that the systems presented in this work are research prototypes and should not be deployed in high-stakes decision-making contexts without careful human oversight. Any future deployment of AFD systems should be accompanied by transparency about model limitations, mechanisms for human review, and continuous evaluation for fairness and bias.

Despite these risks, we believe that this work contributes to raising awareness of fallacious reasoning and can help users recognize misleading arguments. Ultimately, the goal is to support digital literacy and improve resilience against misinformation for both content creators and consumers.

5.5 Conclusion

This thesis has explored the task of automatic fallacy detection through the lens of the FadeIT shared task, achieving competitive results while uncovering fundamental limitations in current approaches. Our work demonstrates that multi-task learning can yield strong performance on task-specific benchmarks, but also reveals that such performance does not necessarily reflect genuine understanding of fallacious reasoning and limitations of the dataset and evaluation system.

Moving forward, the field must place greater emphasis on data quality and diversity, with larger and more carefully annotated datasets that capture a broader range of perspectives, domains, and topics. Models should be trained across multiple domains to learn more generalizable representations, rather than optimizing for a single benchmark. In addition, cross-dataset evaluation should become standard practice, with transparent reporting of both micro- and macro-F1 scores and per-class performance to enable more meaningful and honest comparisons.

Fallacy detection is not just a technical challenge to be optimized away. It is a deeply interdisciplinary problem that sits at the intersection of logic, linguistics, psychology, and argumentation theory. As such, progress will require sustained dialogue between computational methods and theoretical insights to really unlock the full potential of state-of-the-art models.

This thesis represents one step in that ongoing journey. The challenge ahead is substantial, but so is the potential impact: systems that help people recognize fallacious reasoning could contribute to more informed public discourse, better decision-making, and ultimately, a healthier democracy. A goal that is worth pursuing.

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Appendix A

Fallacy Definitions

Table A.1: Fallacies with definitions and examples from the guidelines used by the annotators, span highlighted in bold [4].

Fallacy	Definition and Example
Ad-hominem (AH)	Definition: A personal attack to an individual or a group which deviates from the main thesis. Example: **Gli accoglienti e umanitari francesi non vogliono più i #migranti della #SeaWatch3. La loro parola vale meno di una scoreggia.** ***(The welcoming and humanitarian French no longer want the #migrants on the #SeaWatch3. Their word is worth less than a fart.)***
Appeal-to-authority (AA)	Definition: The author appeals to an authority or a group consensus to support their thesis, without further evidence. Example: **Lo dice anche la #Bundesbank**, i #migranti servono ad abbassare i salari di tutti. *(**The #Bundesbank also says it**, #migrants serve to lower everyone's wages.)*

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Appeal-to-emotion (AE)	Definition: It involves the use of negative or positive personal emotions (e.g., shame, indignation, pity, or affection) to intentionally or unintentionally influence the audience. Example: La manifestazione a #Voghera è **arrabbiata, dignitosa e ordinata**. Sta girando per tutta la città con molti immigrati e l'appoggio della sinistra. **Vuole giustizia**. *(The demonstration in #Voghera is* ***angry, dignified, and orderly****. It is moving through the city with many immigrants and the support of the left.* ***It wants justice****.)*
Causal-oversimplification (CO)	Definition: It involves a simplified and fallacious causal relation. Example: **Ho fatto lavare l'auto. Dopo mesi di siccità, sono sicuro che pioverà presto.** ***(I had the car washed. After months of drought, I'm sure it will rain soon.)***
Cherry-picking (CP)	Definition: Choosing evidence to support a thesis while ignoring any contrary evidence. Example: **Questo video rinfrescherà la memoria a chi l'ha corta e mostrerà ai giovani che Roma ha sempre avuto voragini e alberi caduti dopo forti piogge.** ***(This video will refresh the memory of those with short memories and reveal to the young that Rome has always had sinkholes and fallen trees after heavy rains.)***
Circular-reasoning (CR)	Definition: An error of circularity: the end of an argument comes back to the beginning without having proven itself. Example: **Le leggi sull'immigrazione sono già così fasciste che quando i fascisti promettono di introdurre nuove misure, stanno solo descrivendo le norme fasciste che già esistono e sono applicate da decenni.** ***(Immigration laws are already so fascist that when fascists promise to introduce new measures, they're actually just describing the fascist rules that already exist and have been applied for decades.)***

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Doubt (DO)	Definition: Used to intentionally question the credibility of someone or something. Example: Nella strage di Lampedusa morirono 368 migranti. L'Italia e l'Europa dissero "mai più". **Forse ci credevano davvero?** *(In the Lampedusa massacre, 368 migrants died. Italy and Europe said "never again". **Maybe they really believed it?**)*
Evading-the-burden-of-proof (EP)	Definition: A thesis is advanced without any support as if it was self-evident, meaning that one or more arguments are missing. Example: Trovo geniali **tedeschi e svedesi che censurano i crimini degli immigrati per restare in testa alla classifica dei popoli meno xenofobi.** *(I find it ingenious how **Germans and Swedes censor immigrants' crimes to stay ahead in the ranking of the least xenophobic nations.**)*
False-analogy (FA)	Definition: Two different things are placed on the same level because they are supposed to share similar aspects. Example: Non vi fa sorgere qualche dubbio **uno stato che blocca le #navidacrociera per prevenire il #covid ma fa sbarcare centinaia di immigrati dalle navi delle #ONG e dai #barchini?** *(Doesn't it raise doubts abou**t a state that blocks #cruiseships to prevent #covid, but allows hundreds of immigrants to disembark from #ONG ships and #smallboats?**)*
False-dilemma (FD)	Definition: Presents only two options when there are many. Example: **Queste elezioni saranno uno spartiacque tra chi mette al centro la sostenibilità ambientale e chi difende il combustibile fossile.** ***(These elections will be a turning point between those who prioritize environmental sustainability and those who defend fossil fuels.)***

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Flag-waving (FW)	Definition: Intentionally plays on a sense of belonging to a country, group, or ideology to support an argument. Example: **Con la forza e l'unità di questo popolo orgoglioso** chiediamo a tutta la #Toscana di schierarsi subito. **Vinciamo, insieme, per la Toscana più forte e unita!** *(**With the strength and unity of these proud people**, we ask all of #Tuscany to stand with us immediately. **Let's win, together, for a stronger and more united Tuscany!**)*
Hasty-generalization (HG)	Definition: Generalization drawn from a sample too small or unrepresentative. Example: Se Draghi ha mentito agli Italiani sul vaccino, **allora sta mentendo anche su guerra, clima, transizione ecologica, pnrr, energia, benzine e bollette!** *(If Draghi lied to Italians about the vaccine, **then he is also lying about the war, climate, ecological transition, pnrr, energy, petrol, and bills!**)*
Loaded-language (LL)	Definition: Using words or phrases with strong emotional implications to influence the audience. Example: Epidemia colposa. **Bomba** in arrivo dalla Procura, **valanga** di indagati: **tremano** Speranza e Lorenzin. *(Culpable epidemic. **Bombshell** coming from the Public Prosecutor's Office, **avalanche** of suspects: Speranza and Lorenzin **tremble**.)*
Name-calling-or-labelling (NC)	Definition: Labeling someone positively or negatively to influence the audience. Example: **Un idiota fasciorazzista** condivide una foto di un giovane immigrato inventando fosse senza biglietto; foto condivisa 80mila volte con frasi **razziste**; smentita di Trenitalia: "Aveva solo sbagliato posto". Siete solo dei **razzisti di merda**. *(**A fasci-racist idiot** shares a photo of a young immigrant claiming he was ticketless; photo shared 80k times with **racist** comments; Trenitalia's denial: "He just sat in the wrong seat". You're just **shitty racists**.)*

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Red-herring (RH)	Definition: The argument diverts attention to irrelevant issues. Example: **Siccome pandemia, crisi energetica, crisi climatica, inflazione e tensioni tra Russia e Ucraina non sono sufficienti**, oggi il Corriere ci regala questo articolo "Fidati dei professionisti dell'informazione". *(**Since pandemic, energy crisis, climate crisis, inflation, and tensions between Russia and Ukraine aren't enough**, today the Corriere gives us this article "Trust the professionals of information".)*
Slippery-slope (SS)	Definition: Implies an exaggerated consequence could result from an action. Example: [USER] dà la colpa delle violenze di Peschiera agli italiani. **Un altro vergognoso gradino verso la voragine della stampa italiana**. *([USER] blames Italians for the violence in Peschiera. **Another shameful step towards the abyss of Italian press**.)*
Slogan (SL)	Definition: Brief and striking phrase used to provoke excitement. Example: La difesa dei confini e la lotta all'immigrazione incontrollata di massa rimarranno una priorità. **Difendiamo i confini! #BloccoNavaleSubito** *(The defense of borders and the fight against uncontrolled mass immigration will remain a priority. **Let's defend the borders! #NavalBlockadeNow**)*
Strawman (ST)	Definition: Distorting someone else's argument and then tearing it down. Example: **L'incipit del TG1: "Complice il bel tempo, continuano gli sbarchi a Lampedusa". Vogliamo il mare a forza 8 così ci pensa lui a smaltire i migranti?!** ***(The incipit of TG1: "Thanks to the good weather, landings in Lampedusa continue". Do we want a force 8 sea so it can get rid of migrants?!)***

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Thought-terminating-cliches (TC)	Definition: Short and generic phrase discouraging critical thought. Example: Nella UE tornano i muri antimigranti. **Era ora.** *(In EU, anti-migrant walls come back. **It was about time.**)*
Vagueness (VA)	Definition: Ambiguous words are shifted in meaning or left vague. Example: Ogni centesimo speso in armi è tolto a sanità, ricerca, istruzione e transizione energetica: perciò non sono pro-Putin, che in questo **ha fatto una scelta ben precisa.** *(Every cent spent on weapons is taken away from healthcare, research, education, and energy transition: therefore, I am not pro-Putin, who **has made a very clear choice** in this regard.)*